



Ninaivu

நினைவு · memory

Your family's photographs, at home.

The application guide · version 0.1.0

Ninaivu
the library

Mugil
backup

Sudar
the photo studio

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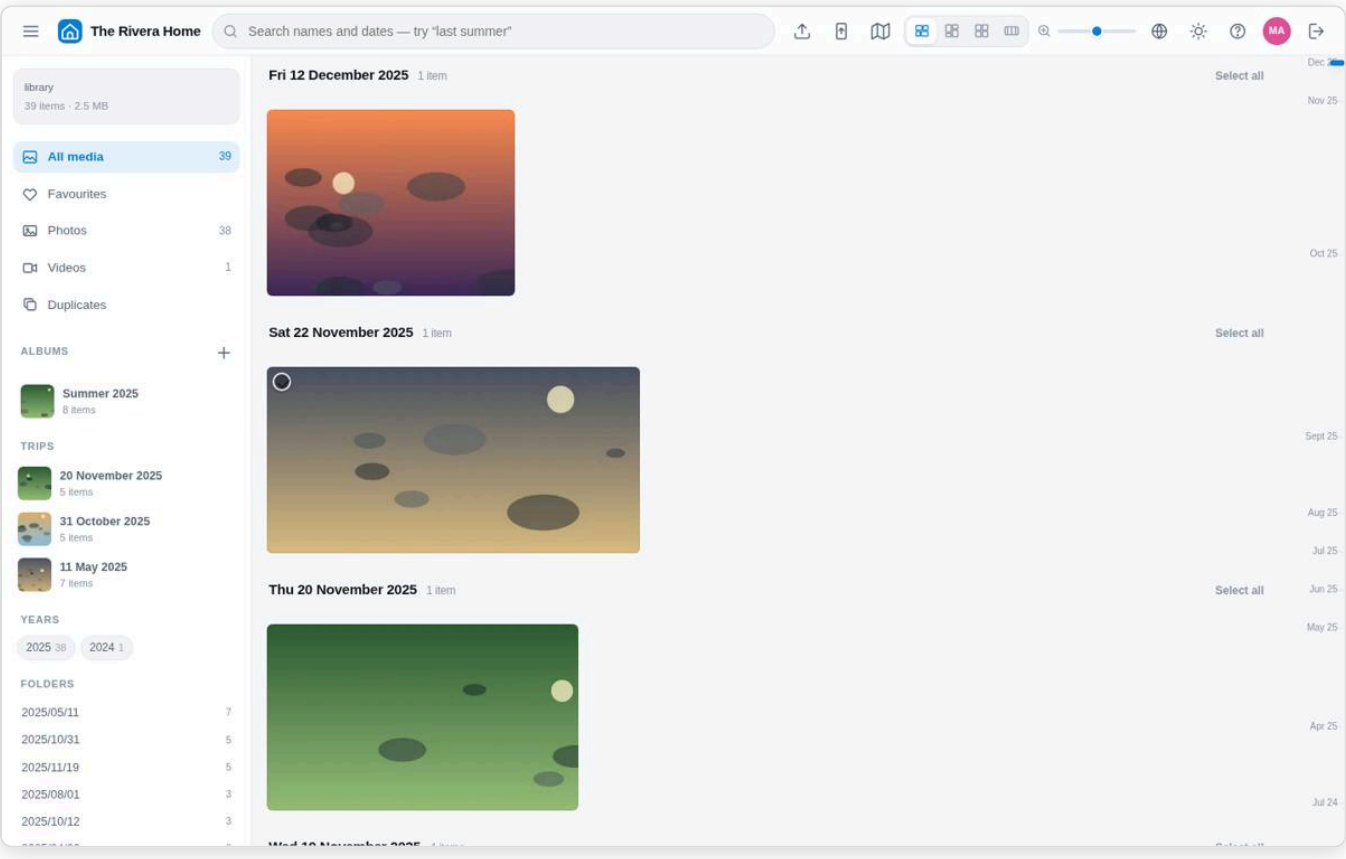
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The pictures use a generated sample library, not anybody's photographs. Everything here runs on a computer you own; nothing leaves the house unless you choose where it goes.

Ninaivu

நினைவு · *memory* — your family's photographs, at home.

Ninaivu is a private photo and video library for one household. It runs on a computer you own, indexes what is on your own disks, and gives your family what people usually leave to a cloud service: faces and names, search by description, places on a map, "on this day", a link for the grandparents, and phones that send their photographs home. Nothing leaves the house unless you choose where it goes — and the backup goes encrypted.



It has three parts, each with a name:

Ninaivu	The library: gallery, faces, search, places, albums, sharing, and the three roles
Mugil (முகில், cloud)	Backup: an encrypted copy of the library on Google Drive, and the household's phones sending their photographs to the house
Sudar (சூடர், glow)	The photo studio: light, colour, straightening, crops and suggestions, in the browser, never changing the original

Where to start

1. Install — Windows, macOS, Docker, or a checkout. Ten minutes.
2. The first day — the folder, the household, what the scan works out by itself.
3. Family and roles — who sees what.
4. Backup — a copy outside the house, tested every week.
5. Remote access — reaching it from away.
6. AI — what runs on your machine, and what a small machine skips.
7. Troubleshooting.

Then **the guide**, everything each screen does: the family app for the household, the console for whoever runs it. Every screen, pictured: a tour of the screens.

Status

Ninaivu 0.1.0 is the general edition of a server that has run one household's library for a year. It has some 3,500 tests. The roadmap says what 1.0 still needs.

Install

Ninaivu runs on a computer that stays on: a desktop under the stairs, a mini PC, an old laptop, a Mac mini, a NAS that runs Docker. It needs Python 3.12 or newer when run from a checkout; the installers bring their own.

Windows

Download `Ninaivu-<version>-windows-x64.exe` from the latest release and run it. It installs for your account only (no administrator prompt), puts a private Python and everything Ninaivu needs under `%LOCALAPPDATA%\Programs\Ninaivu`, and adds **Ninaivu** to the Start menu. Leave *Start Ninaivu at sign-in* ticked and it is always there.

If the installer is not signed (the release notes say), Windows SmartScreen says *Windows protected your PC*: choose **More info** → **Run anyway**. Once Ninaivu is in the winget catalogue, `winget install Ninaivu.Ninaivu` will work too.

Opening **Ninaivu** puts an icon in the system tray. Its menu starts and stops the server and opens the family app and the console. More on the tray.

macOS

Download the `.dmg` for your Mac — `arm64` for Apple silicon, `x86_64` for Intel — from the latest release, open it and drag **Ninaivu** to Applications.

If the app is not notarised (the release notes say), macOS will not open it the first time. Try once, then go to **System Settings** → **Privacy & Security** and choose **Open Anyway**; after that it opens normally. Once Ninaivu is in Homebrew, `brew install --cask ninaivu` will work too.

Ninaivu lives in the menu bar, not the Dock. Its menu starts and stops the server, opens the family app and the console, and can start at sign-in.

Docker

For a NAS or a Linux box:

```
git clone https://github.com/javajaga-usa/Ninaivu.git
cd Ninaivu
cp installers/docker/.env.example installers/docker/.env # set MEDIA_DIR to your
photo folder
docker compose -f installers/docker/docker-compose.yml up -d
```

The family app answers on port 5000 to the whole network; the console on 3000, published on the NAS itself only (`127.0.0.1:3000`). To make the administrator from another computer, open the console through an SSH tunnel or on the NAS, and give the **setup code** printed in `docker logs ninaivu`. The `.env` file is only for folders and ports: everything about how Ninaivu behaves is chosen in the console, and nothing in `.env` undoes it at the next restart. The image leaves out search by description unless built with `--build-arg WITH_AI=1`. The operations guide covers systemd, reverse proxies, HTTPS and large libraries.

From a checkout

```
git clone https://github.com/javajaga-usa/Ninaivu.git
cd Ninaivu
start.cmd ~/Pictures           # Windows: or double-click start.cmd
sh launcher/start.sh ~/Pictures # macOS and Linux
```

The launcher makes a virtual environment, installs what is missing (nothing large: search by description is added later, if you want it), serves HTTPS on port 443 — or 8080 on Linux, where an ordinary user may not use 443 — finds free ports and opens your browser. `python -m ninaivu <folder>` runs the server directly, on plain HTTP and port 80 (8080 without the right to use 80), unless given `--https`.

On Windows, `start.cmd` needs no Python beforehand: when it finds no Python 3.12 or newer, it installs Python 3.12 for the current user first (with winget, or from python.org when winget is missing), then carries on. No administrator rights are needed.

On a Mac, double-click **Setup Ninaivu.command** in the Ninaivu folder once instead of using a terminal. It clears the download quarantine, installs Python 3.12 if the Mac has no Python 3.12 or newer (after asking: with Homebrew when the Mac has it, otherwise the installer from python.org, which is run only when it carries the Python Software Foundation's signature, and for which macOS asks your password), installs what Ninaivu needs, and puts two apps in Applications: **Ninaivu**, which starts the library in a Terminal window and asks once which folder holds your photos, and the **Ninaivu Control Panel**, which starts and stops it without one. `sh launcher/start.sh` offers the same Python install. On Windows, **Start - Ninaivu Control Panel.vbs** opens the Control Panel once `start.cmd` has run. More on the Control Panel.

What you see first

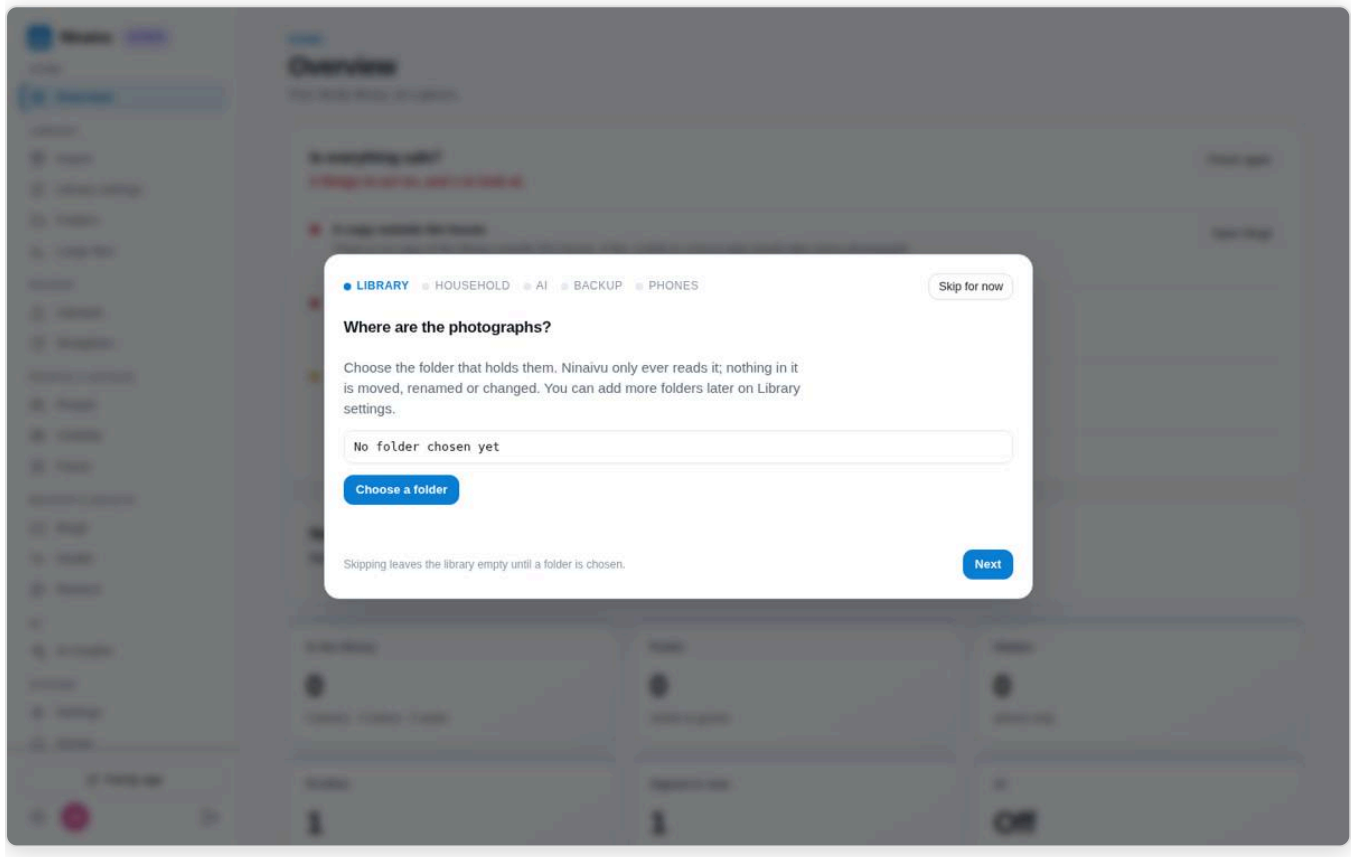
The first screen makes the administrator. On the computer Ninaivu runs on, that is all; from any other device it also asks for the **setup code** printed where Ninaivu started (the terminal, the log, or `docker logs`), so nobody else at home can claim a new library first. After that the console walks you through the first day.

Two addresses

Ninaivu has two faces on two ports: the **family app**, where everyone looks at the library, and the **console**, where the administrator runs it. The console answers on this computer only until you open it to the home network on the Server page; the family app is on the home network.

The first day

Right after the administrator is made, the console walks through the five things a new library needs. Each step can be skipped and each is just a page of the console, so nothing here is a second way of doing anything.



1. Where are the photographs?

Choose the folder that holds them. Ninaivu only ever reads it: nothing is moved, renamed or changed. You can add more folders later on **Library settings**, and sweep old drives and memory cards into one dated archive with **Import** — which also understands a Google Photos Takeout export, dates, places, descriptions and albums included.

2. Who is in the household?

A **family member** taps their name and enters a PIN. A **guest** signs in with a password (or a PIN, or a tap, set on **People** later) and sees only what you make public. Add the people you know of now; the rest can come later on **People**. A profile without a PIN can be opened from any phone or

computer on your home network, so give a PIN to anyone whose photographs should stay theirs. More on roles.

3. What should Ninaivu work out by itself?

Everything here runs on this computer; no photograph leaves it. **Find the people in photographs** needs a 37 MB model, downloaded when you turn it on. **Name the places** needs an 11 MB list of places, downloaded once from GeoNames. **Read the words in photographs** needs the text reader from Extras. **Search by description** — finding "the beach at sunset" without anybody tagging it — is about 2 GB, added only if you press *Add search by description*; it is worth it on a computer with 8 GB of memory or a graphics card. Each pass takes time on a large library and can be turned off part-way. More on AI.

4. A copy outside the house

A fire, a theft or a failing disk would take every photograph. **Mugil** keeps an encrypted copy on Google Drive and checks once a week that it restores. It needs a Google account and an encryption key — nothing is uploaded until the key is made — which take a few minutes on the Mugil page; the Overview reminds you until it is done. More on backup.

5. On the household's phones

Open the address shown on a phone on the home Wi-Fi and add it to the home screen: it opens like an app. From its backup screen, the photographs and videos you choose are sent to the house; keep the page open while it sends.

After that

The console's **Overview** answers two questions before anything else: *is everything safe?* and *what needs you?* — uploads to approve, faces without a name, photographs that may be on their side. Come back to it now and then.

Family and roles

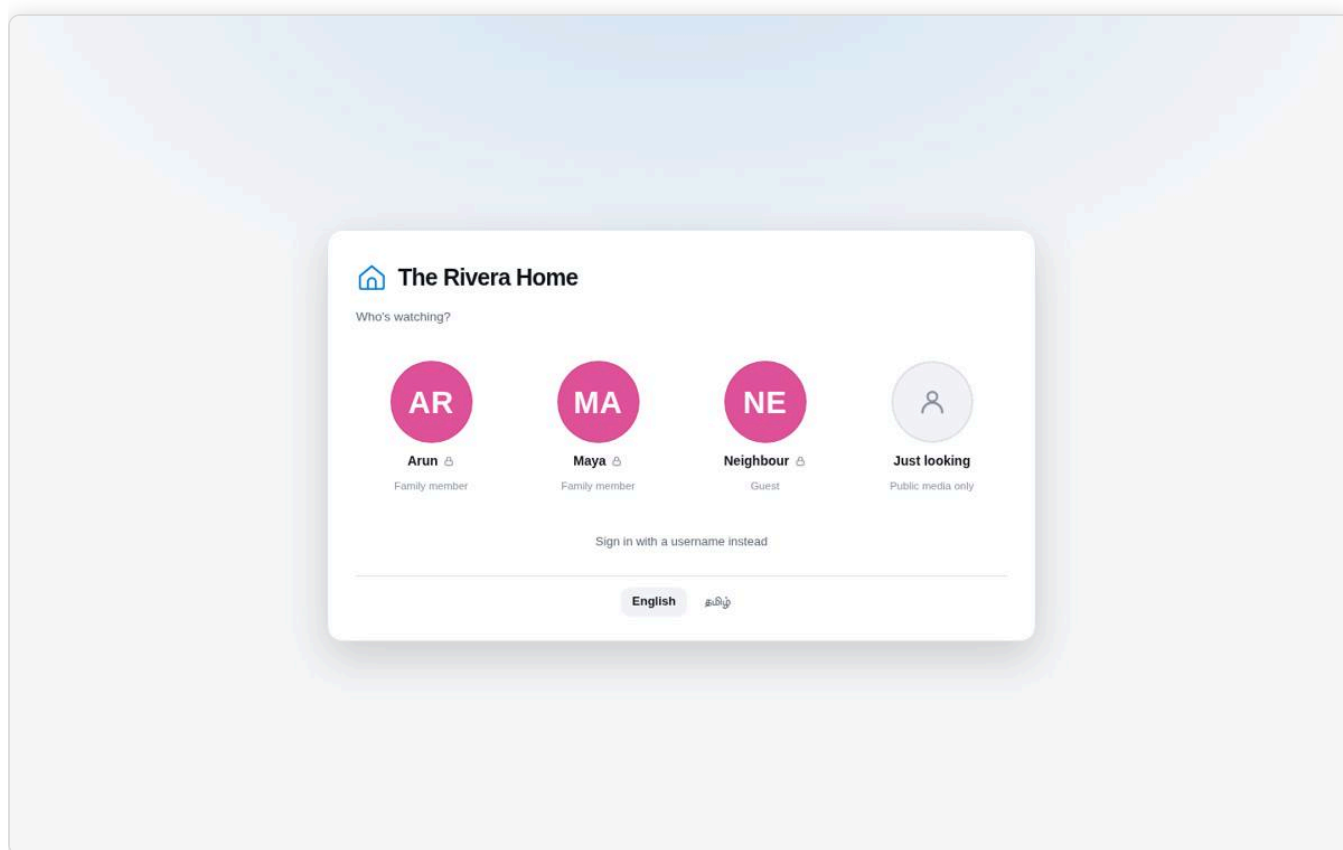
Three roles, and every photograph has a visibility. That model is the product; everything else is built on it.

Role	Signs in with	Sees
Administrator	a password, on the console (and in the family app, where the same account can also delete and change visibility)	everything, and runs the server
Family member	a PIN, a password, or a tap, from the profile picker	everything marked for the family
Guest	a password, a PIN, or a tap	only what was chosen for them

A profile set to *tap to enter* needs no secret, which suits a shared tablet — and means any device on the home network can open it. Give a PIN to anyone whose photographs should stay theirs.

With **open browsing** on, a visitor can look at what is public without signing in at all; off, the whole library is private. Screenshots, documents and photographs of screens are admins-only by default (*hide screens*), and so is every sound recording.

Both apps lock after 15 minutes without use; the person's PIN or password opens them again, and whatever was running carries on meanwhile.



Visibility

Every folder has a setting — everyone, family, or admins only — and every photograph can be set on its own. **People & access** → **Visibility** holds the rules for everyone first (whether visitors may browse, whether explicit content is screened, how far back in time each role may look) and then each folder's own setting. Every change can be undone in one step.

A family member can also be given a folder of their own: a private corner of the library that only they and the administrator see.

Sharing

Family members and administrators can make a link for one photograph or an album, with an optional password, scoped to what its maker could already see. Guests cannot share.

Faces and names

People & access → **Faces** shows the people the face matcher found, grouped. Name one face and the rest of its group follow. Faces are found and matched on this computer. The names are part of the index, so they travel only where the index does: into Mugil's daily copy, encrypted with your key.

Uploads

Photographs the family send from their phones wait under **Review** → **Uploads** for the administrator to file or refuse. A family member's edit in Sudar waits the same way. Nothing a family member does changes the library without the administrator seeing it.

Language

Ninaivu is in English and Tamil. Each person chooses on the language button (the sign-in and lock screens have it too), and the choice is saved on their profile, so it follows them to every device. In Tamil, the name is நினைவு.

Dates

A role can be limited to a window of time — guests see this year's photographs, say, and nothing older. Date access.

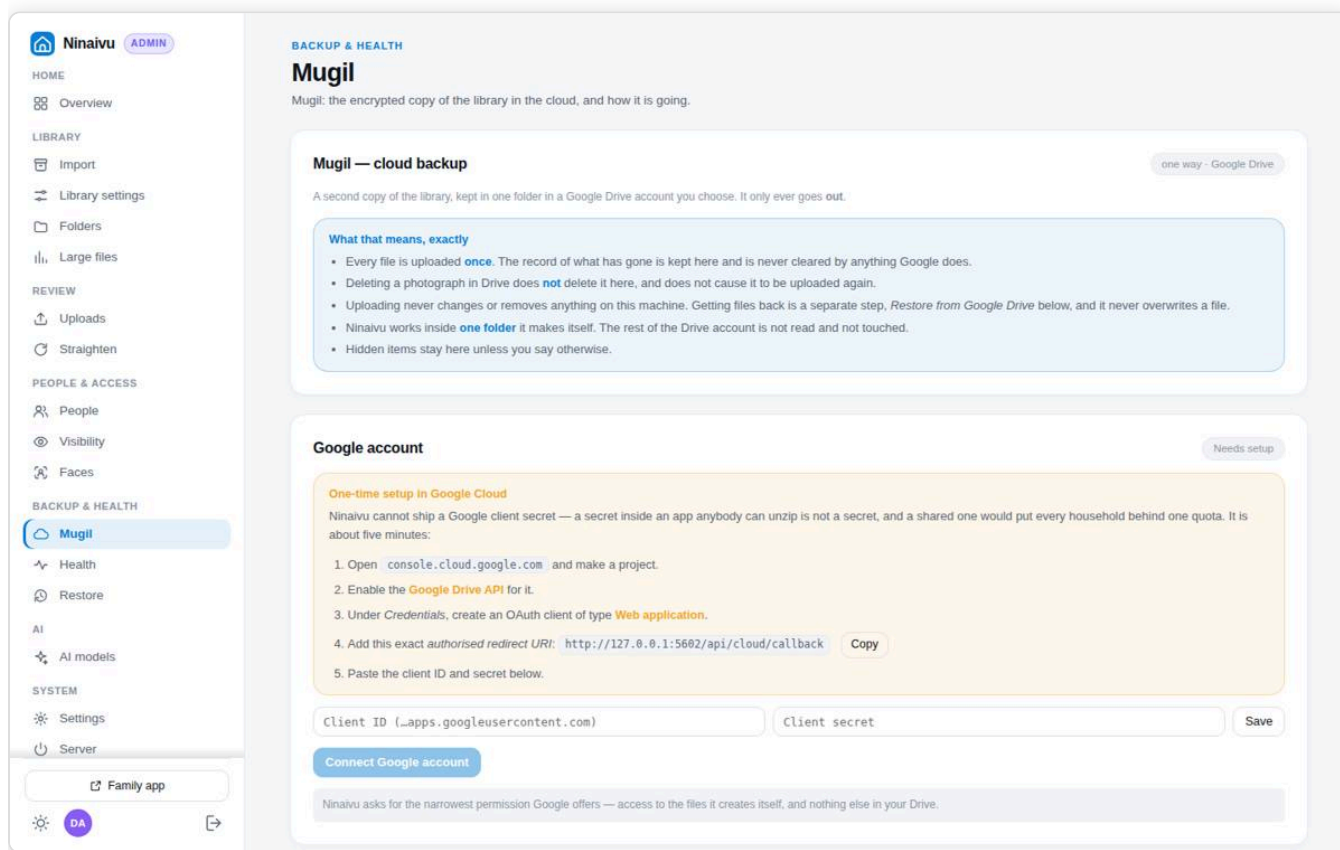
Backup

Three things can lose a library: the disk, the house, and a mistake. Ninaivu keeps a copy against each.

Mugil — a copy outside the house

Mugil (முகில், *cloud*) keeps an encrypted copy of the library on Google Drive. It is one way: nothing in Drive can change, move or remove anything at home, and deleting a photograph from Drive does not delete it here.

Encryption is on from the start, and **nothing is uploaded until you have made the encryption key** on the Mugil page — so no photograph, and no copy of the index with the names given to faces, ever goes up unencrypted. An administrator can switch encryption off; the page then says plainly that photographs and names go up as they are.



On the **Mugil** page: the Google account (a few minutes, once), the encryption key and its **recovery file**, what is sent and what is kept back, the speed and the hours it may use, and **Test restores** — the weekly test that brings a few files back and checks they match, with its history.

The recovery file

The copy in Drive is useless without the key. Ninaivu writes a recovery file and asks you to keep it somewhere that is not this computer — a USB stick in a drawer, a password manager. Without it, a lost computer means a lost library, however good the copy in Drive is. What is in it.

Google Drive is the only destination today; a second disk, a NAS path, S3 and WebDAV are on the roadmap.

Restore

Backup & health → **Restore** brings photographs back from the cloud copy in three steps: what to bring back, where to put it, and the key. Nothing here ever overwrites a file that is already there.

The index

The index — faces and the names given to them, albums, who may see what — is copied on this computer once a day, seven copies kept (**Health** shows them), and into Drive with the library, encrypted with the same key, so a lost computer loses nothing but time. To put a copy back: stop Ninaivu and run `ninaivu restore <file>`. Moving to another machine.

Phones

On a phone, the family app's backup screen sends the photographs and videos you choose to the house over the home Wi-Fi. A web page cannot read the phone's library by itself or keep going in the background, so you pick what to send and keep the page open while it sends; it skips what is already here and carries on where it stopped. What arrives waits under **Review** → **Uploads** for the administrator, unless the administrator switched on **trust phone backups**; an administrator's own phone skips the review.

Is everything safe?

The console's Overview runs the household's safety checks every time it opens — a copy outside the house, a backup that restores, copies of the index, healthy drives — and says what to do about each.

Remote access

At home, Ninaivu answers to a name on the home network (`ninaivu.local`). Away from home, it is reached through something the household set up — and Ninaivu treats *how* as a choice, made once on **System** → **Server** → **Away from home**.

Choice	What it is	What it needs
Tailscale	A private network between your own devices; the easiest	Tailscale installed on the server and on each phone
WireGuard	The same idea, set up by hand	the tunnel, and its subnets typed here
A tunnel	Cloudflare Tunnel or the like: a public name that reaches the house	the tunnel, and the name typed here
Your own reverse proxy	Caddy or nginx in front, with a certificate	the proxy (examples)
Nothing	Home network only	—

The default works out Tailscale by itself when it is on the machine.

Each choice says what it still needs, decides which addresses count as *away from home*, which addresses are listed for the household, and whose certificate is served.

HTTPS

Ninaivu makes its own certificate authority and serves HTTPS with it. Each device that should not see a warning trusts that certificate once; the tray and the console's Server page have the button, and the tray page has the details for Windows and macOS. Tailscale and a reverse proxy bring their own certificates.

The console from other devices

The family app answers on the whole home network; the **console answers on this computer only** until you tick *Open this console from other devices at home too* on the Server page (Ninaivu restarts). `--admin-host` sets its address outright.

Everything Ninaivu sends out on its own

Remote access is a way *in* to the house. Faces, names and photographs are never sent anywhere by it. This is the complete list of what Ninaivu asks of the internet without a person pressing something at that moment:

What	When	How to stop it
A request to GitHub's releases page: is there a newer version? Nothing about this computer is in it.	Once a day	Server → the update check
Mugil: the photographs and the daily copy of the index, encrypted, to the Google account you connected	While Mugil is on	Mugil
Tailscale's HTTPS certificate for this computer, through Tailscale	With HTTPS on and Tailscale installed, every few weeks. The name <code><computer>.<tailnet>.ts.net</code> then appears in public certificate logs, as every Let's Encrypt name does.	All settings → <code>tailnet_https</code>
The image model's weights, from Hugging Face	Once, at the first start after search by description was added; nothing after that	Do not add search by description
A model download that stopped half-way	At the next start, until it finishes	AI models
The list of places, from GeoNames	Once, at the first scan after <i>Name the places</i> was switched on	Leave it off
Map tiles, from OpenStreetMap — they show roughly where your photographs were taken	Whenever someone opens the map, if <code>map_tiles</code> is on (off by default; the map uses a built-in outline otherwise)	All settings → <code>map_tiles</code>

Model downloads, notifications by email or webhook, and extensions such as Gemini happen only once an administrator has set them up.

AI

Everything on this page runs on your own computer, and no photograph, face or name is sent anywhere by it. The one exception is an extension you switch on that says otherwise on its switch (Gemini, below). The models themselves are downloaded once, when you turn their feature on.

Two kinds of computer

Ninaivu is tested on two kinds of machine, and works out which yours is when it starts (**System** → **Performance** says which, and why):

	Basic	Full
The machine	anything else: a mini PC, an old laptop, a single-board computer, a desktop without a graphics card or without the image model	a graphics processor with at least 6 GB of memory, or Apple silicon, or any computer with 8 GB of memory once search by description is added
The gallery, albums, sharing, roles	yes	yes
Faces, grouped and named	yes	yes
Sideways photographs put right	yes	yes
Places named from the location	yes	yes
Text read from photographs (with the text reader)	yes, slowly	yes
Search by words in names, dates, places and people	yes	yes
Mugil, the encrypted copy	yes	yes
Tags, "the beach at sunset" search and descriptions from the image model	—	yes
Finding visually similar photographs (S in the viewer)	—	yes
Videos described from several moments	—	yes
Creative Studio: generative edits, object removal, upscaling	—	with a graphics card

On a Basic machine, *auto* means the light engine, so it never tries to load the large models and starts quickly. Choosing the image model outright (`--ai clip`, or `ai_engine` on **All settings**) still loads it. Setting `hardware_tier` on **All settings** overrides the measurement.

Adding search by description

It is not installed by default: it is about 2 GB, plus the model itself. Add it from the first-day walk-through or **Settings** → **Extras** → **Search by description**, then restart Ninaivu. On a Full machine it is used from then on; the model's weights are fetched once at that first start, and after that Ninaivu tells the model libraries to make no request at all (`HF_HUB_OFFLINE`). The Docker image leaves it out unless built with `WITH_AI=1` .

What each pass does

AI → **AI models** has a switch for each, saying which model it needs and whether that model is here.

- **Faces** — a 37 MB detector and matcher. Groups are named on **Faces**; name one and the rest follow.
- **Places** — an 11 MB list of places, downloaded once from GeoNames; the photograph's own location becomes "Paris" or "the beach". The map uses a built-in outline of the world; street-map tiles from OpenStreetMap are off unless `map_tiles` is switched on, because each tile asked for shows roughly where a photograph was taken.
- **Sideways photographs** — a 77 MB model that says which way up a photograph goes. By default it looks by itself after every scan and turns what it finds straight away, as one batch that **Undo** on **Review** → **Straighten** puts back; switch that off and what it finds waits there for approval.
- **Text in photographs** — the text reader from **Extras**; slow on a processor.
- **The image model** (Full, once added) — CLIP or SigLIP: tags, "the beach at sunset" search, descriptions, visually similar photographs.
- **Videos** (Full) — described from several moments rather than the poster frame.

Sudar

Sudar (சூடர், *glow*) edits in the browser: light, colour, straightening, crops, looks, and suggestions measured from the picture. "Make it warmer" in plain words. The original is never changed; a copy is saved beside it, and so can any result a tool makes — a cut-out, a coloured print — a family member's copy waiting for the administrator's approval like an upload. The **Photo Studio** (*Edit photo* in the viewer) renders through the same engine, so a slider means the same thing in both; the family guide has every panel.

Its **Skin** and **Hair** tools work on every person in the photograph, each judged against their own skin, using the face model from **AI models**. They leave a bindi, sindoor or sacred ash exactly as it is, and have no setting that lightens a complexion. Hair the colour of what is behind it is found with the help of *Background removal (RMBG-1.4)* when that model is installed, and painted in by hand otherwise; **Add hair** draws new hair from the strands beside it, on this machine. See the family guide.

Its **Enhance tools** each need a model, and the card says which:

- **Upscale** — *Upscale ×4 (Real-ESRGAN)*, from **AI models**.
- **Restore faces** — *Restore faces (GFPGAN 1.4)*, from **AI models**.
- **Colourise** — a DDColor ONNX model. The catalogue cannot pin one yet, so this is the one model placed by hand: put `ddcolor.onnx` in the `onnx` subfolder of the models folder (**AI models** shows the folder), or point `NINAIVU_COLORIZE_MODEL` at the file. With Restore faces too, **Revive old photo** does both in one tap.

Each runs on this machine and nothing leaves it; with the Creative Studio extension, an AI server workflow can take any of them instead.

Extensions

Anything that cannot keep the core's two promises — nothing leaves the house unless you chose where, and it runs on a small machine — is an extension, off until switched on under **System** → **Settings** → **Extensions** (it takes effect when Ninaivu next starts), and its switch says what it does:

- **Creative Studio** — generative edits and the heavy tools with large models, on this computer with a GPU or on a ComfyUI server at home. The AI server page refuses an address on the internet, so nothing leaves the house (a Tailscale address counts as home).
- **Gemini** — edits and descriptions from Google's models. A re-encoded copy of the photograph goes to Google when, and only when, a request names it. Only an administrator can send one, unless *Let family members send photographs to extensions that go outside the house* is ticked under Settings → Extensions.

More on extensions.

Troubleshooting

The phone cannot open Ninaivu

- The phone must be on the home Wi-Fi, not mobile data.
- Use the address the console's **Server** page lists under *for the household*. `localhost` only works on the server itself.
- **Network access** on the Server page must be on; off, Ninaivu answers on the server only. Turning it on needs a restart.
- Windows: if the firewall blocks it, the startup log prints the two `netsh` commands that open the ports; run them once in a terminal opened as administrator.

I cannot open the console from my laptop

The console answers on the computer Ninaivu runs on only, until you tick *Open this console from other devices at home too* on the Server page (at that computer). Ninaivu restarts, and the console is then on the home network too.

The browser warns about the certificate

Ninaivu serves HTTPS with its own certificate authority. Trust it once per device: the tray's **Trust the HTTPS certificate**, or the Server page's button, and the tray page for doing it by hand on Windows and macOS. Or put Tailscale or a reverse proxy in front, which bring their own.

Nothing appears after choosing the folder

The library is indexed in the background; a large one takes a while. The Overview's **Library** numbers and the Server page's log show it going. If the folder is on a drive that is not plugged in, the gallery says so rather than showing nothing.

The scan is slow

System → **Performance** says what this computer can do and what would help. On a small machine, turn off the passes you do not need on **AI models** (text in photographs and video moments are the expensive ones), or choose *power-saving* on the Server page while the household is using the computer. A scan can be stopped and picks up where it left off.

A model would not download

The models come from GitHub and Hugging Face. Check the machine has internet at the moment of the download; the AI models page says which file failed. Once a model is here nothing is asked of the internet again.

Mugil says the copy did not restore

The weekly test brings a few files back and compares them. If it fails, **Mugil** → **Test restores** says which file and why; the usual causes are a changed Google password (sign in again on Mugil) and a full Drive.

Mugil says nothing is uploading

Encryption is on by default, and nothing is uploaded until the encryption key exists. Make it on the Mugil page and keep the recovery file.

Ninaivu will not start

The tray's **View the log** (or `.ninaivu-control/server.log` beside a checkout) has the reason. The common ones: the port is taken by another program (Ninaivu then moves to the next free one and says so; `--port` chooses another, `--strict-port` refuses to move), the state folder is on a drive that is not mounted, or another Ninaivu is already running from the same state folder.

Moving to another computer

System → **Move to another computer** lists the three things that make an installation and, on the new machine, points the index at wherever the library landed without re-indexing. The details.

Getting help

Issues on GitHub. Say which version (**System** → **Settings** shows it under *This home*), what you did, and what the log said.

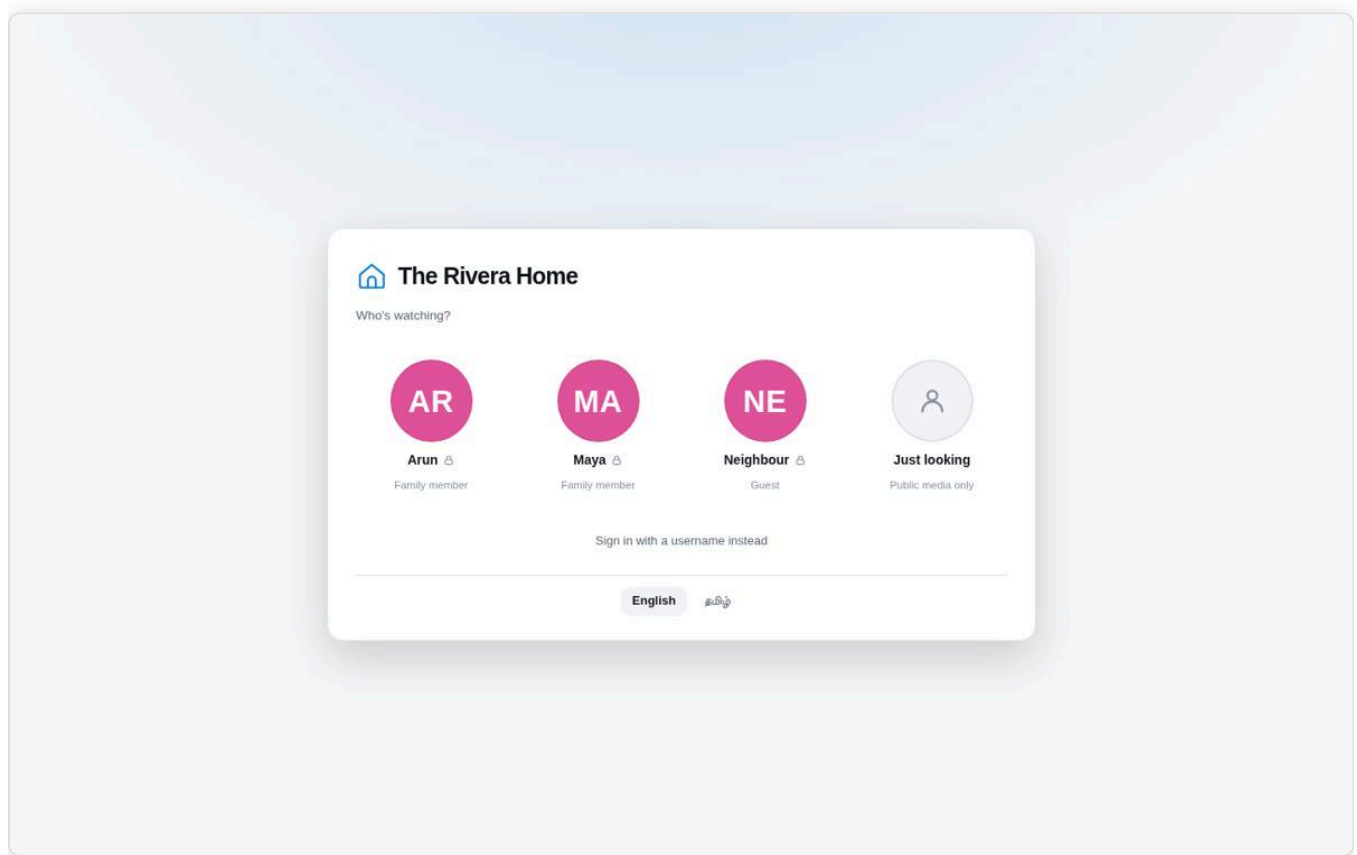
The family app

Everything a family member or guest needs, and nothing else. You do not need to know anything about servers to use Ninaivu: it opens like a website, and on a phone or tablet you add it to the home screen once and it behaves like an app.

Two things explain most of how it behaves:

- **Everything has a visibility.** Every photograph is *public*, *family* or *admins only*. What you see depends on which profile you came in as.
- **The management screens are somewhere else.** The administrator runs Ninaivu from a separate console on its own address. A family member or guest can change nothing about the library from the family app; an administrator signed in there can also delete and change visibility.

Coming in



The family app opens on the household's profiles. Tap yours. What that costs depends on how the administrator set it up:

Tap to enter nothing at all — right for the family tablet or a shared television

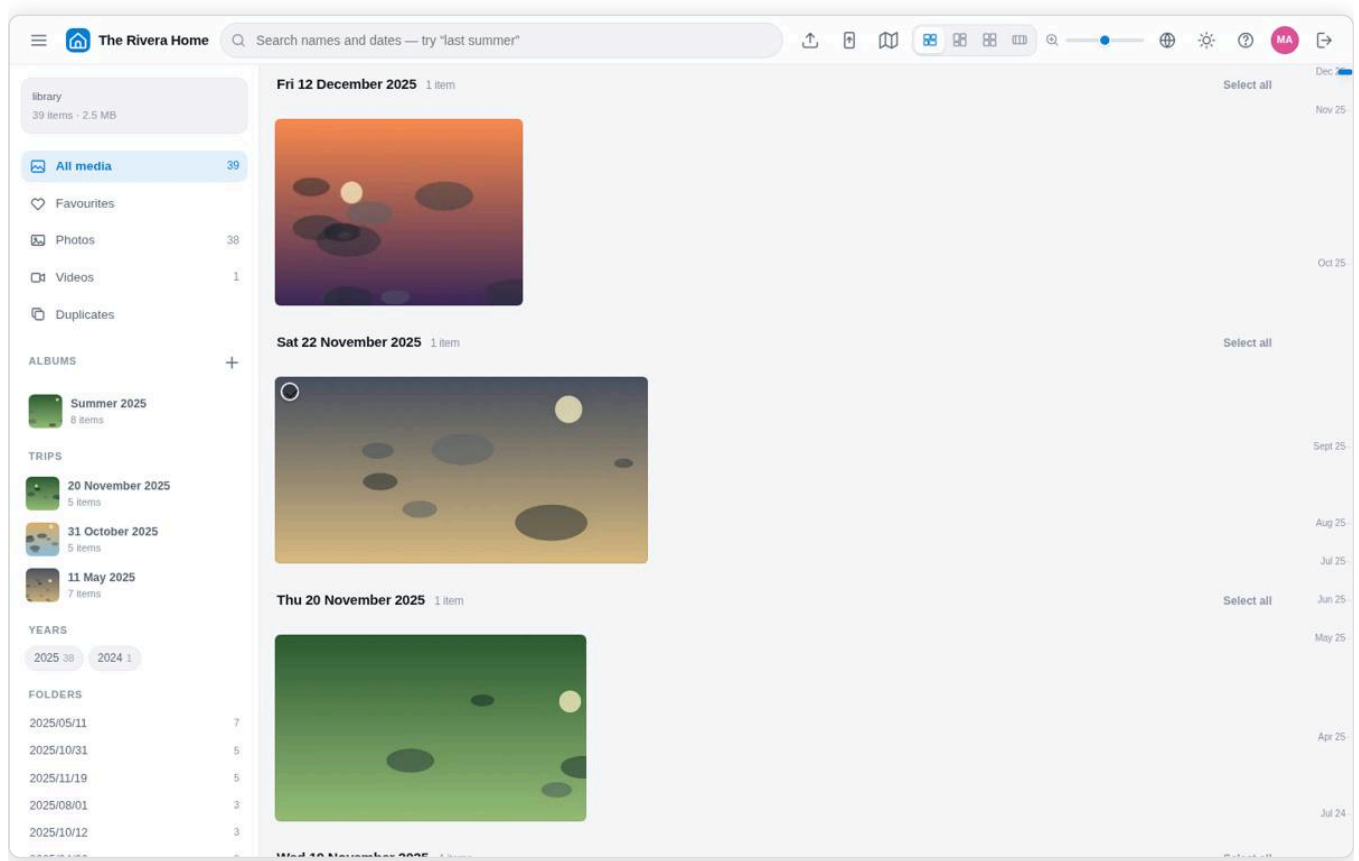
PIN four to eight digits, shown as a small lock on your tile

Password the full form, for guests and anyone who wants it

Just looking with open browsing on, a tile that needs no profile and shows only what is public

Administrators never appear on this picker; they sign in on the console. You can change your own name, colour and picture whatever your role.

The gallery



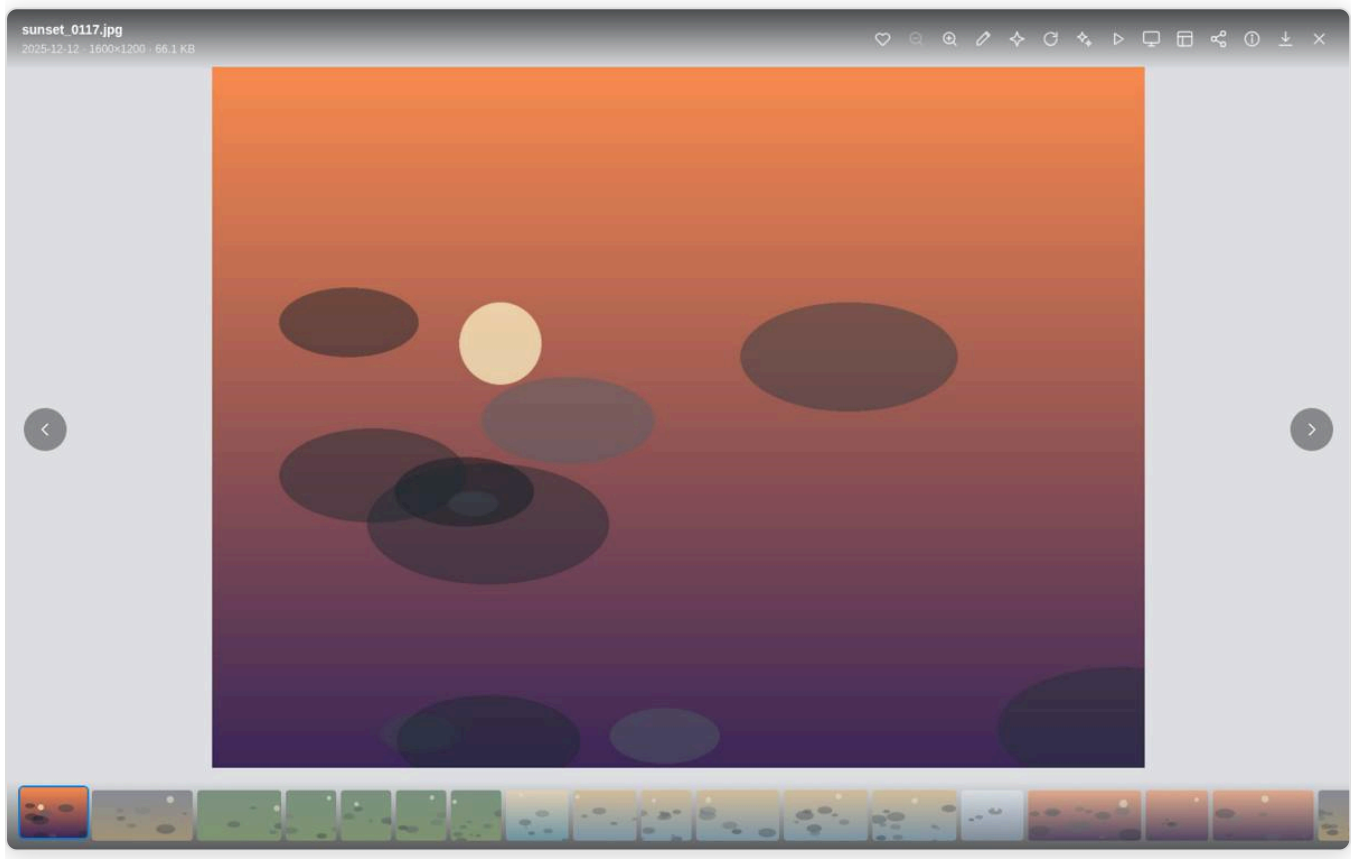
Everything you may see, newest first, grouped by day. Four layouts, and they are a real preference: **justified** suits mixed shapes, **masonry** tall photographs, **grid** scanning, **filmstrip** one long roll. **+** and **-** change the thumbnail size; **T** switches the theme.

The left column is the library's shape: photos, videos, favourites, duplicates the index found, albums, trips the dates suggest, then years and folders. Click any of them and it becomes a removable filter chip; chips stack, so *beach* + *2019* + a person is one click each. The strip down the right edge jumps anywhere in the library by month, as smoothly on a hundred thousand photographs as on a hundred.

Search takes plain words. On a Full machine (AI) the image model ranks results by how well the picture matches — "the beach at sunset" works as written, nobody tagged anything. On a Basic machine search covers names, dates, places, people and the words in file and folder names. The map button shows every located photograph grouped by place.

Selecting. Click the circle on a tile, Shift-click for a range, **X** on the focused tile, or **⌘/Ctrl+A** for everything shown. With a selection you can favourite, download or add to an album at once; an administrator also changes visibility in bulk.

A photograph



The whole photograph fitted to your screen, with its date, place and people, and the actions your role may take.

- **Zoom** with the wheel or a pinch, drag to pan, double-click to zoom to a point; **0** puts it back. Arrow keys or a swipe move on.
- **I** — **details**: camera, lens, ISO, aperture, where the date came from, the location if the file carried one.
- **S** — **find similar**: up to twelve photographs that look like this one — the other attempts at the same moment. It needs search by description (AI); without it the panel says the photograph is not analysed yet.
- **F** — **favourite**. Favourites are private to you, even when several people share the same folder. Guests cannot favourite.

- **R** — **rotate** (Shift for the other way). A turn saved by a family member is saved for the whole household; the file itself is not touched unless an administrator writes the turn into it.
- **D** — download the original (family members and administrators).
- **Share link** — a link for this photograph, with an optional password, that shows exactly what you could already see.
- **Edit photo** — the Photo Studio, below (family members and administrators). The button beside it opens **Sudar**, the AI studio, below that.
- **Space** — a slideshow. **K** — the **Ambient Frame**: full screen, a slow drift across each picture, a clock in the corner, a new photograph every few seconds. It is the reason to keep an old tablet on the kitchen wall. **Esc** comes back.

Live Photos and motion photos play their short clip; Ninaivu pairs the still and the video during indexing.

Press **?** anywhere for the full list of keys.

People

If faces are on, a row of people appears in the sidebar. Click one to see that person's photographs. The count is *yours* — how many you are allowed to open — so two people in the house may see different numbers for the same person. Naming people and confirming matches are the administrator's job on the console; in the family app, People is read-only.

Albums

Any family member can make an album and add to it. Albums hold references, not copies: deleting an album never touches a photograph. An album can be shared as a link, with an optional password.

Sending photographs

Family members can upload from the gallery; the files wait for the administrator under **Review** → **Uploads** before they join the library, and a name that already exists gets a short suffix rather than overwriting anything. On a phone added to the home screen, the backup screen sends the photographs and videos you choose: keep the page open while it sends — a web page cannot read the phone's library by itself or keep going in the background — and it skips what is already here and resumes where it stopped. The limit is 512 MB per upload. Guests cannot upload.

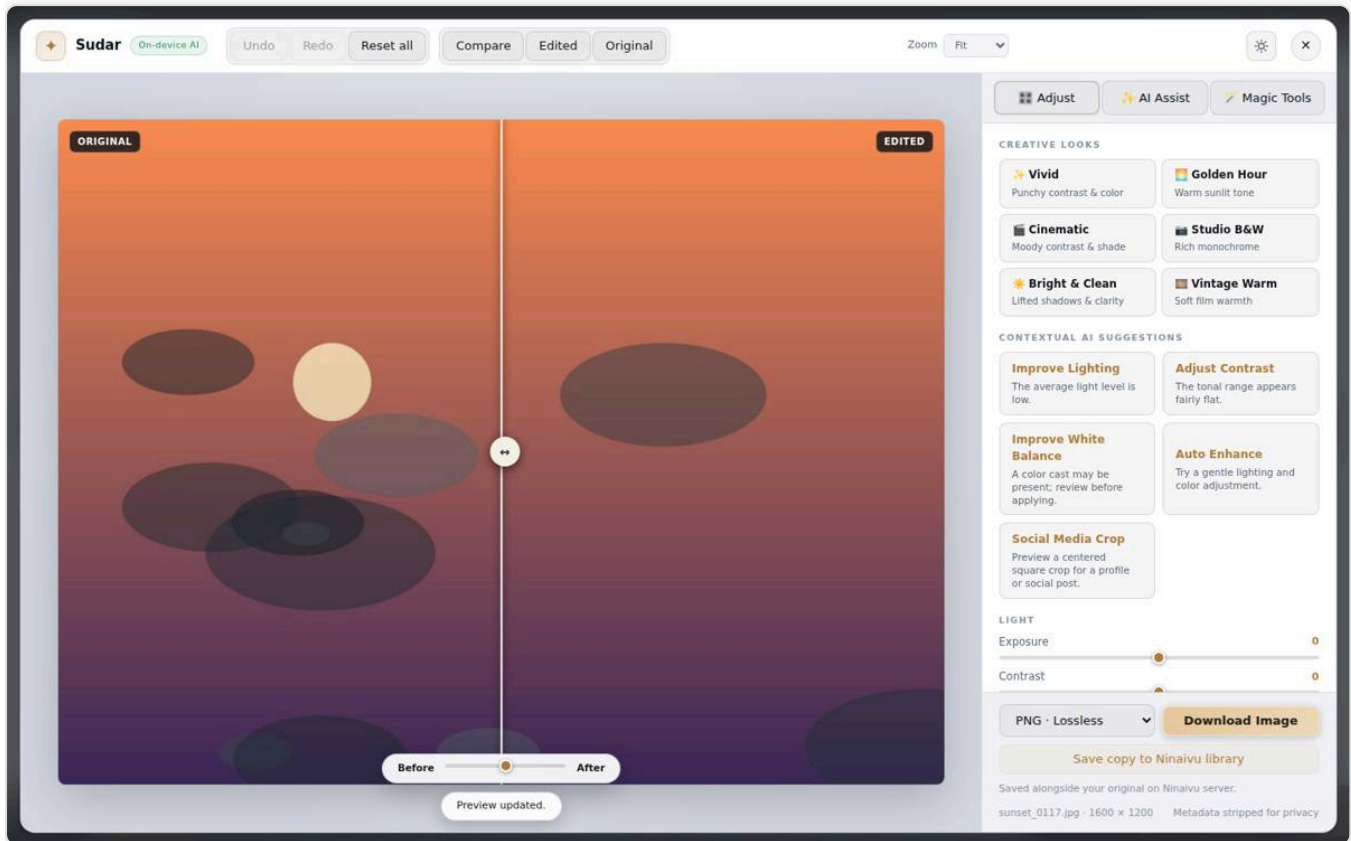
The Photo Studio

Edit photo opens the Photo Studio: light, colour and detail, on this device, with the original kept. Every slider means the same thing as the slider of that name in Sudar, because the two share one engine.

- **Light** — a histogram at the top; exposure (40 on the slider is one stop), contrast, highlights and shadows judged by their neighbourhood so a face in front of a bright window opens up without a pale ring round it, whites, blacks, dehaze; and a **tone curve** drawn over the histogram, for all channels or one, with points added by clicking and removed by dragging them off the square.
- **Colour** — white balance and tint; **vibrance**, which knows where brown skin of every complexion sits and leaves it nearly alone, and saturation; an eight-band **colour mixer** (hue, saturation, luminance); **black and white**, whose tones come from the mixer's luminance sliders; **split toning**; and **Natural skin tone**, which judges the colour of the whole photograph by the skin in it and moves warmth and tint only as far as skin of any complexion looks natural.
- **Detail** — clarity (negative softens), sharpening on lightness only with radius and masking, luminance and colour noise reduction, film grain, and a vignette with midpoint and feather.
- **Skin and Hair** — the people in the photograph, each against their own skin: below.
- **Looks** — ten for family photographs, each shown as a small picture of the photograph being edited and laid over your own sliders with an amount: Natural, Festival, Golden hour, Soft portrait, Backlit rescue, Revive old print, Film, Monsoon, Classic black and white and Warm black and white. None raises the lightness of skin or drains its colour.
- **Crop** — straighten, ratios, flip across and flip upside down. **Export** — full size, or 3840, 2048 or 1080 pixels on the long edge.

The photograph is worked a strip at a time, so a 24-megapixel file fits in the browser's memory. Saving writes a copy in the same folder with what the camera wrote — make, model, lens, exposure settings, place — carried over; the original is never changed. An administrator's copy joins the library at once; a family member's waits under **Review** → **Uploads** for approval.

Sudar, the photo studio



Sudar (சூடர், *glow*) edits a photograph in the browser: looks, suggestions measured from the picture, light and colour sliders (exposure, contrast, shadows, highlights, dehaze; vibrance, saturation, white balance; clarity, sharpness, noise, vignette), straightening and crops (square, 16:9, 4:5, and 9:16 for a story), with the original on the left and the edit on the right. "Make it warmer" in plain words works too. The original file is never changed; a copy can be saved beside it, and a family member's copy waits for the administrator's approval like an upload.

Any result can be saved. A cut-out, a blurred background, a colour pop (the subject in colour, the rest in black and white), an upscaled, restored or colourised picture, a generated preview: each has **Save to library** beside **Download**, and goes the same way as the sliders' edit — published beside the original for an administrator, into the approval queue for a family member. **Object removal** and **Clothing colour** put their result back on the photograph with **Apply to photo**, so the sliders and Save are still there.

Enhance tools — **Upscale**, **Restore faces**, **Colourise** — are always on the card; one that is not set up is greyed out, and the line under it says which model it needs and that an administrator adds it (AI). With both of the last two, **Revive old photo** restores the faces and then colourises an old print in one tap.

With the Creative Studio extension on (AI), Sudar also does generative edits — including a new **hairstyle**, which is drawn rather than adjusted, from ten ideas under the prompt, and **Add hair**, which paints where hair should be and has the AI server draw it — object removal and upscaling, on this computer or on an AI server at home. Nothing leaves the house.

The Creative Studio (the Magic tool of that name, not the extension) makes things from photographs, in the browser: a restoration story and a then-and-now comparison, a family story album with a recorded memory, a postcard, a collage in three layouts, a cinematic slideshow with music, selective colour, a sheet of named versions, a recipe card and a photo calendar. Photographs come from your files or **from the library**; the picture creations come as a landscape, a square or a 9:16 story, and can be **saved to the library** like an edit, or downloaded, or printed. The album, slideshow and comparisons export as a web page that works on its own.

Skin and hair

Open **Skin** or **Hair** in the Photo Studio, or **Skin & hair retouch** in Sudar — they are one set of tools — and Ninaivu looks for every face in the photograph, on this computer, with the face model the People page already uses. Across the top are the people it found: **Everyone**, then each face. Choose one and a slider moves for that person alone (a dot beside it says so); choose Everyone and it moves for everybody who has not been given a setting of their own. **Improve faces** gives each person what stood out about them — a face in shadow is lifted, a yellow or pink cast is turned back towards skin, a shiny forehead is calmed, grey hair is covered — and nothing else. It is one step of Undo, and every slider it moved is still a slider.

Look younger, at the top of the Skin panel, does in one tap what these tools can honestly do about age, for whoever is chosen and by the amount you set: smoother and more even skin, calmer shine, brighter under-eyes, grey covered, strands brought out, thin hair filled. It only moves the sliders below, so each can be changed afterwards; it lightens nothing and redraws nothing, and the person stays exactly who they are. Ninaivu has no age model, on purpose: the ones that exist redraw the face.

What these tools do and do not do:

- **Each person is judged against their own skin.** A fair face and a very dark one in the same photograph are not pushed towards the same colour.
- **A bindi, kumkum, sindoor, sacred ash or sandal paste is left exactly as it is.** Nothing here smooths, recolours or blurs it, and its colour cannot spread into the skin beside it.
- **Nothing makes skin lighter or fairer.** *Face light* is exposure for a face that is in shadow; skin of every colour is lifted by the same amount, so a dark face stays dark. It is only offered when the whites of the person's eyes show the face really is dim — never because the skin is dark.
- **Eyes, brows, lips and teeth are not touched.** *Soften* takes blemishes and leaves pores; a fold, an eyelid or a mole is kept.
- **Hair and beard:** *Strands & shine*, *Fuller hair* (fills thin hair: where light shows through between strands, the gaps are brought down to the strands' tone with their texture kept; it adds no hair where there is none, and never reaches skin), *Cover grey*, and *Hair colour* (black, dark brown, brown, chestnut, henna, burgundy or any colour). When hair cannot be told from what is behind it — black hair against a black wall — Ninaivu asks the background-removal model, if it is installed (AI), where the background is; without it, it says so rather than guess, and you paint the hair in with **Add** and **Remove**, in the Photo Studio and in Sudar alike. What you paint goes with the picture if you crop or turn it.

- **Add hair.** The third brush in the Hair panel: paint where there should be hair — a receding hairline, a thin crown — and it is drawn from the hair beside it, the same strands, colour and shine running on; *Added hair* says how much shows. It is this person's own hair, made on this computer, and it works best for a band of a finger's width or two along the hair that is there. For more than that — a bald patch, a whole new head of hair — Sudar's **Add hair · generative** Magic tool paints the area and has the AI server draw it, when the Creative Studio extension has a *Paint and ask* workflow.
- **Natural skin tone**, in the Photo Studio's **Colour** panel, judges the colour of the whole photograph by the skin in it — indoor light is yellow, a shaded courtyard is blue — and moves the warmth and tint sliders only as far as the edge of what skin of any complexion looks like.

Finding faces needs the face model (**AI models** → **Finding and grouping faces**, part of the essential set). Without it the panel says so, and the brush still works.

Language

The language button switches between English and Tamil (the sign-in and lock screens have it too). It is saved on your profile, so every device you use follows it.

The screen lock

After 15 minutes without use, the app locks; your PIN or password opens it again, and anything in progress carries on meanwhile.

Sound recordings

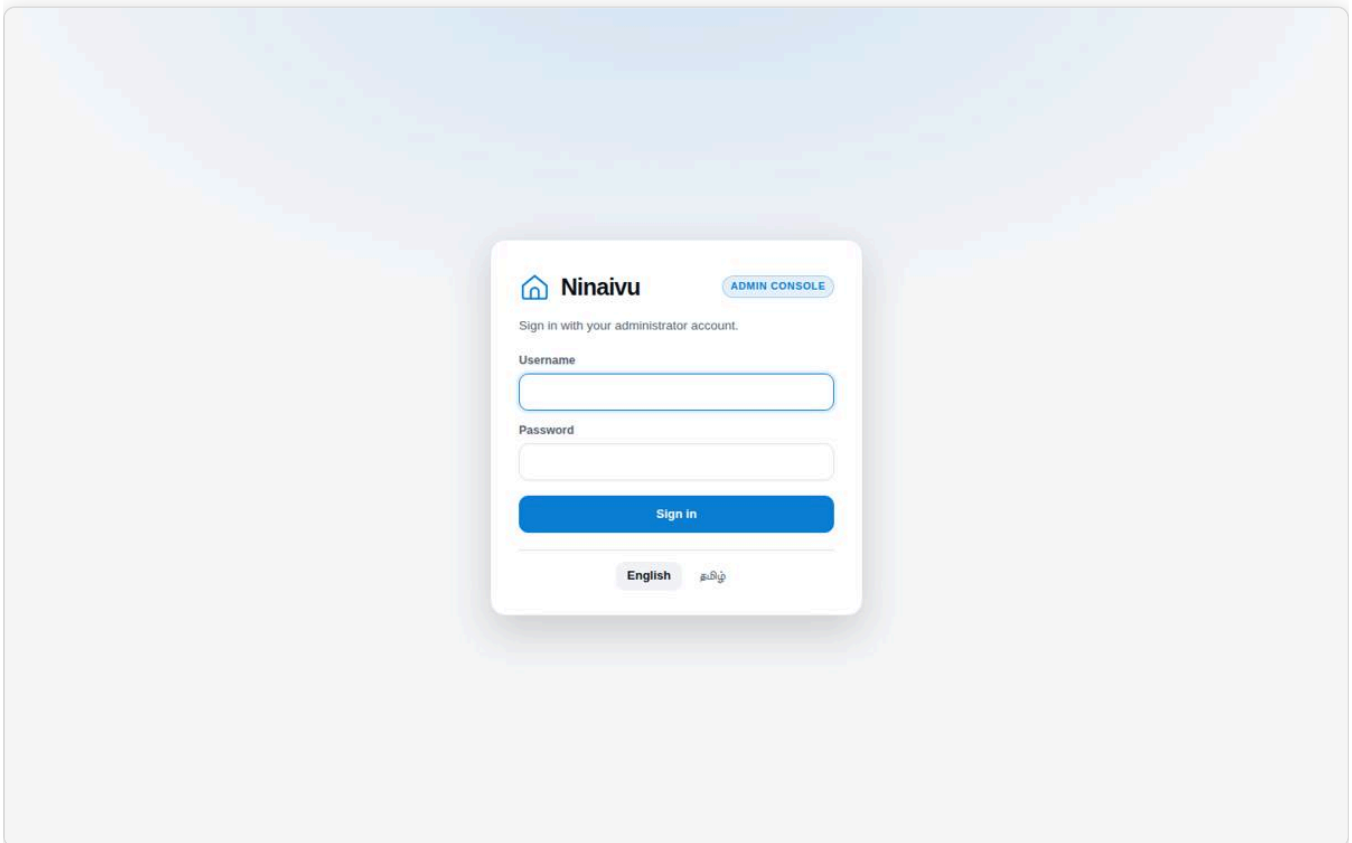
Sound files in the library — voice notes, voicemail, dictation swept up from old drives — are administrator-only, always. A photograph is looked at on purpose; a recording plays out loud the moment somebody taps it, in a room with other people in it.

On a phone

Open the address the administrator gives you while on the home Wi-Fi, then **Add to Home Screen**. It opens like an app and works offline for what it has already shown. Away from home, it works only if the household set up remote access.

The console

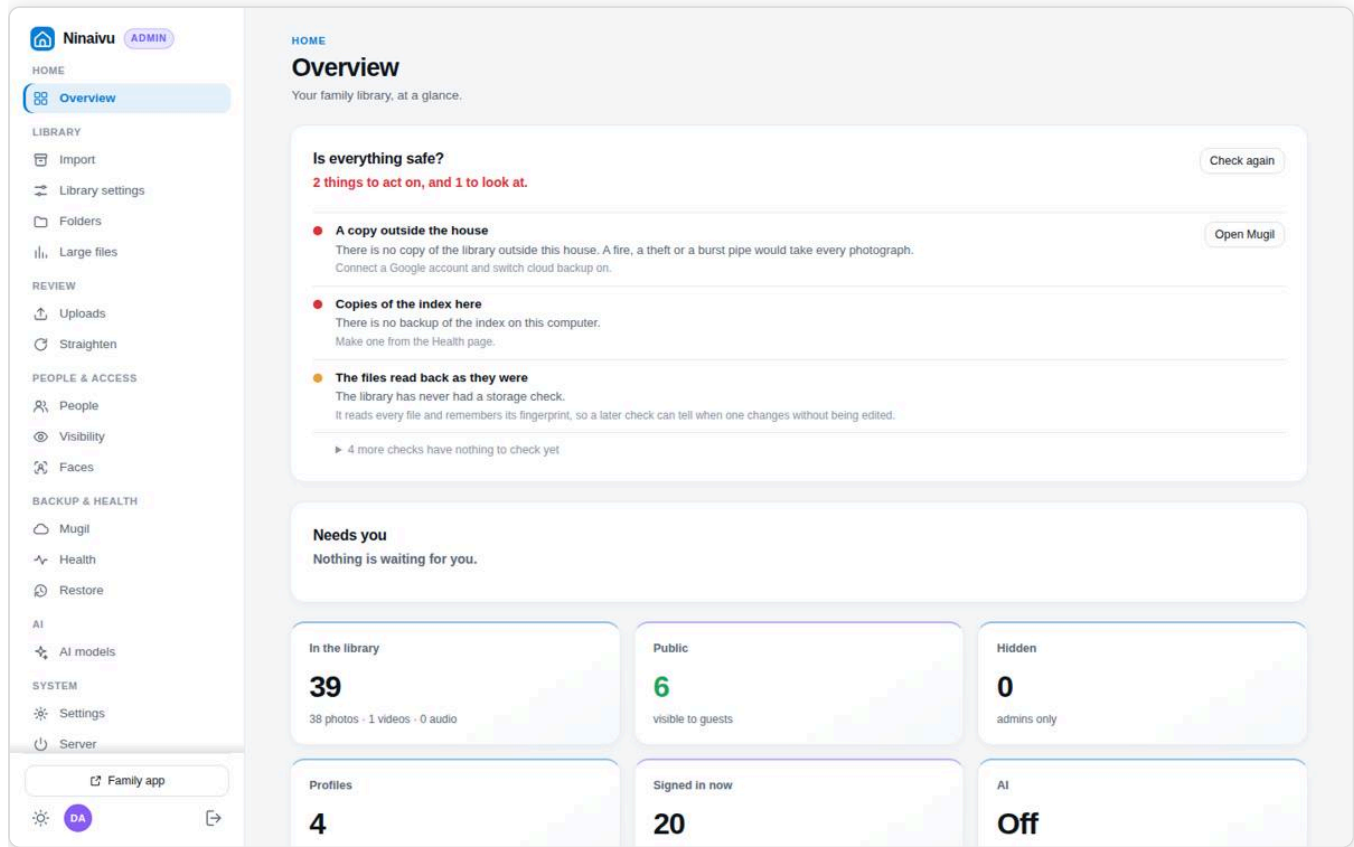
Everything the administrator does: getting a library indexed, deciding who sees what, and the tools — Import, Faces, Mugil, Health. The console answers on its own port, on the computer Ninaivu runs on only until you open it to the home network on the Server page, and always takes a password, never a PIN.



Two apps, two ports

The family app and the console are two applications in one process. The console's routes are *absent* from the family app, not merely forbidden: people, visibility rules, Import, Mugil and settings exist only on the console port. An administrator signed in to the family app can still delete photographs and change their visibility from the gallery. A session made on the family app cannot open the console, even an administrator's.

Home — Overview



The first screen answers two questions before anything else. **Is everything safe?** runs the household's safety checks — a copy outside the house, a backup that restores, copies of the index, healthy drives — and says what to do about each. **Needs you** counts what is waiting for a person: uploads to approve, faces without a name, photographs that may be on their side, things that went wrong. Then the numbers, and what each role would actually see. The Overview is also where the first day opens, once.

Library

Import

LIBRARY

Import

Bring your memories together in one organized archive.

Consolidate drives into one archive
Sweeps photos, video and audio off however many old drives, cards and backup folders you have into one archive filed as YYYY/MM/DD. Sources are opened read-only and **nothing is ever deleted** — every copy is re-read from disk and SHA-256 matched before it counts.

1 Source drives and folders
Every subfolder is scanned, however deep. Each folder gets its own Photos / Video / Audio switches.

No sources yet. Add a folder below — a drive letter is fine, and every folder beneath it is included.

e.g. D:\OldPhotos or /Volumes/Backup Browse... Add source

SCAN EVERY FOLDER FOR ☒ Photos ☒ Video ☒ Audio Apply to all

2 Archive destination
Where the consolidated archive is built. Pick the archive's own root — choose a folder inside one and Ninaivu uses the root instead, and says so.

e.g. D:\Photo Archive Browse...

☒ Deep scan — identify files by content, not just by extension

3 Run

Start consolidation Dry run

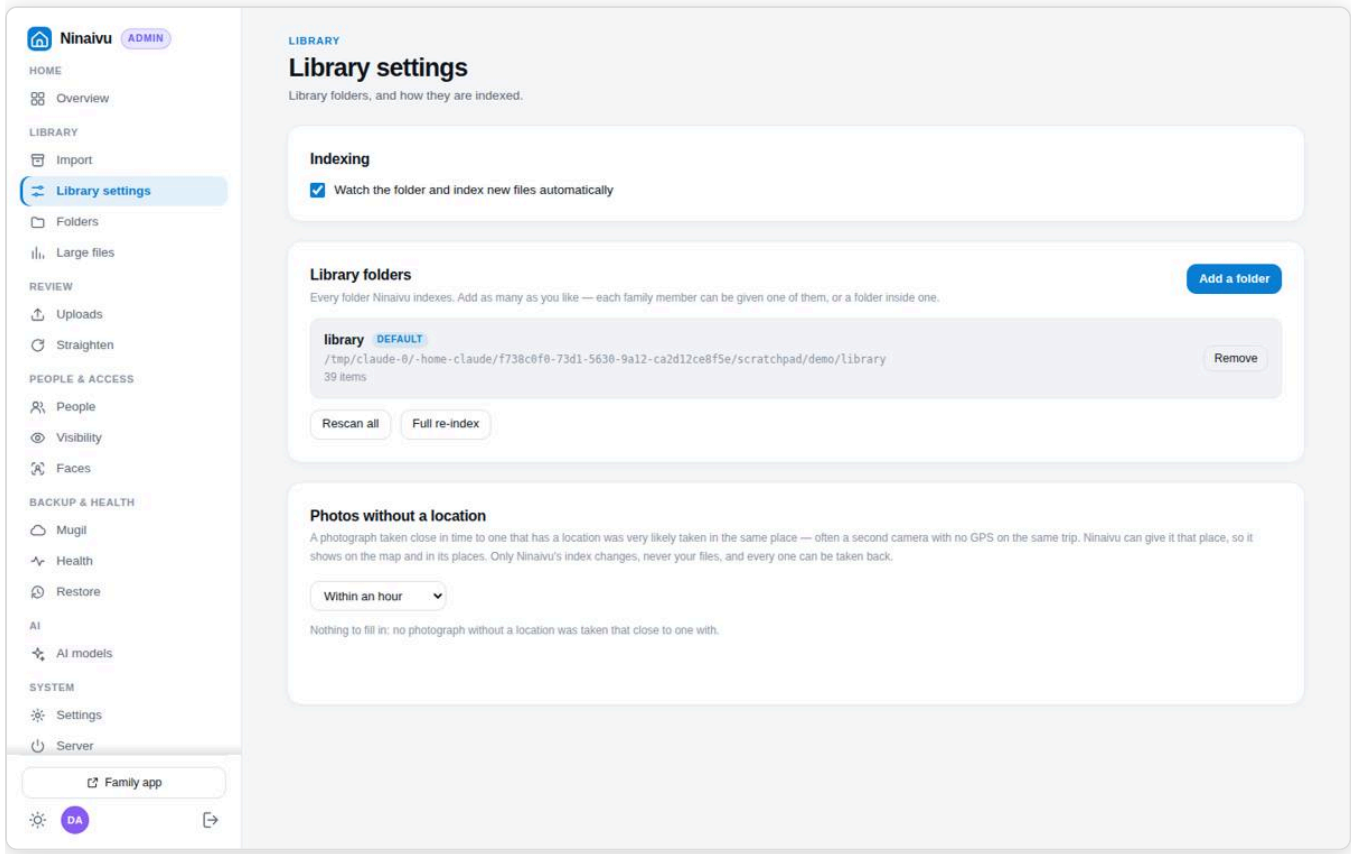
Audit archive Retry failed Export manifest Recovery report Reset report

Start also resumes — an interrupted run picks up exactly where it stopped. Dry run decides everything and writes nothing. Audit re-reads every archived file and re-checks its hash.

The first Library page on purpose: a library usually begins with old drives, memory cards and backup folders, swept into one archive laid out as YYYY/MM/DD . Every file is hashed on the way in and read back to verify the copy; duplicates are recognised by content and left in place, logged; sources are only ever read. **Dry run** decides everything and writes nothing. **Start consolidation** also resumes: an interrupted run picks up where it stopped. Files under 60 KB — icons, thumbnails — are left alone.

When the run has finished the page offers to **add the archive to the library**. A Google Photos Takeout export comes across whole: its sidecars give the dates, places and descriptions, and once the archive is indexed the page offers to **make its albums** from the copies it kept. The **Archive Guardian** re-reads a rotating sample of archived files every day and says when one has changed.

Library settings

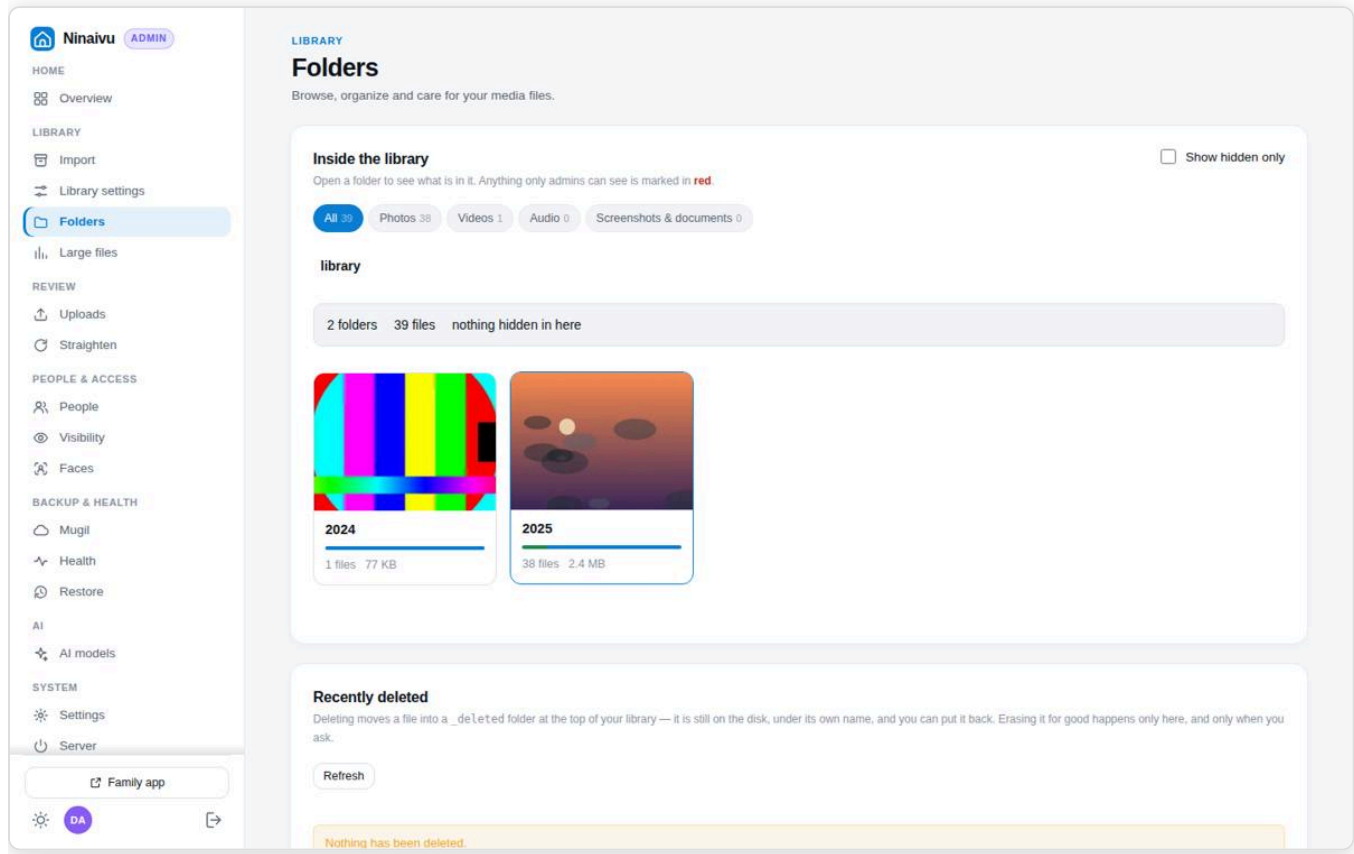


Which folders make up the library, and how they are indexed: whether Ninaivu watches them for changes, whether hidden files are indexed, what the scan works out by itself. Add as many folders as you like; a folder on a drive that is unplugged stays a library folder and the gallery says so rather than showing nothing.

Where the dates come from: the camera's own date first, then the container's metadata, then a Takeout sidecar, then a date in the file name, then a YYYY/MM/DD folder path, and only then the file's own creation or modification time. The viewer's details panel says which one was used.

When a library folder cannot be written — an NTFS drive on a Mac, say — edited copies, approved uploads and phone backups go to ~/Pictures/Ninaivu instead, which joins the library on first use.

Folders



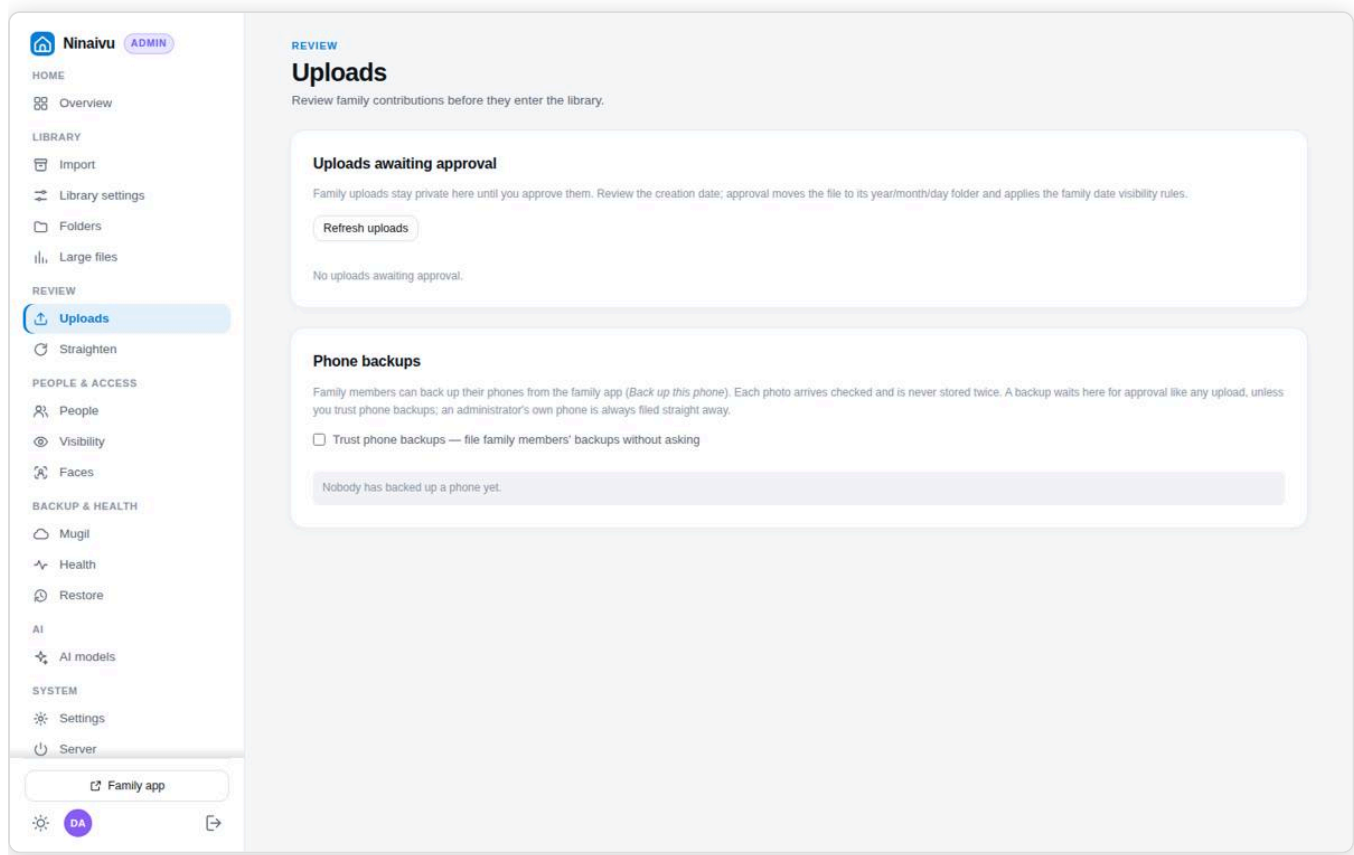
The library as folders, with what is in each and each folder's visibility, and the **recycle bin**. Deleting in Ninaivu moves a file into a `_deleted` folder beside the library, where it can be put back with its faces and albums intact. Nothing is erased for good unless you empty the bin here (it asks first) or set a number of days on All settings.

Large files

The files taking the most space, to look over once and mark as reviewed.

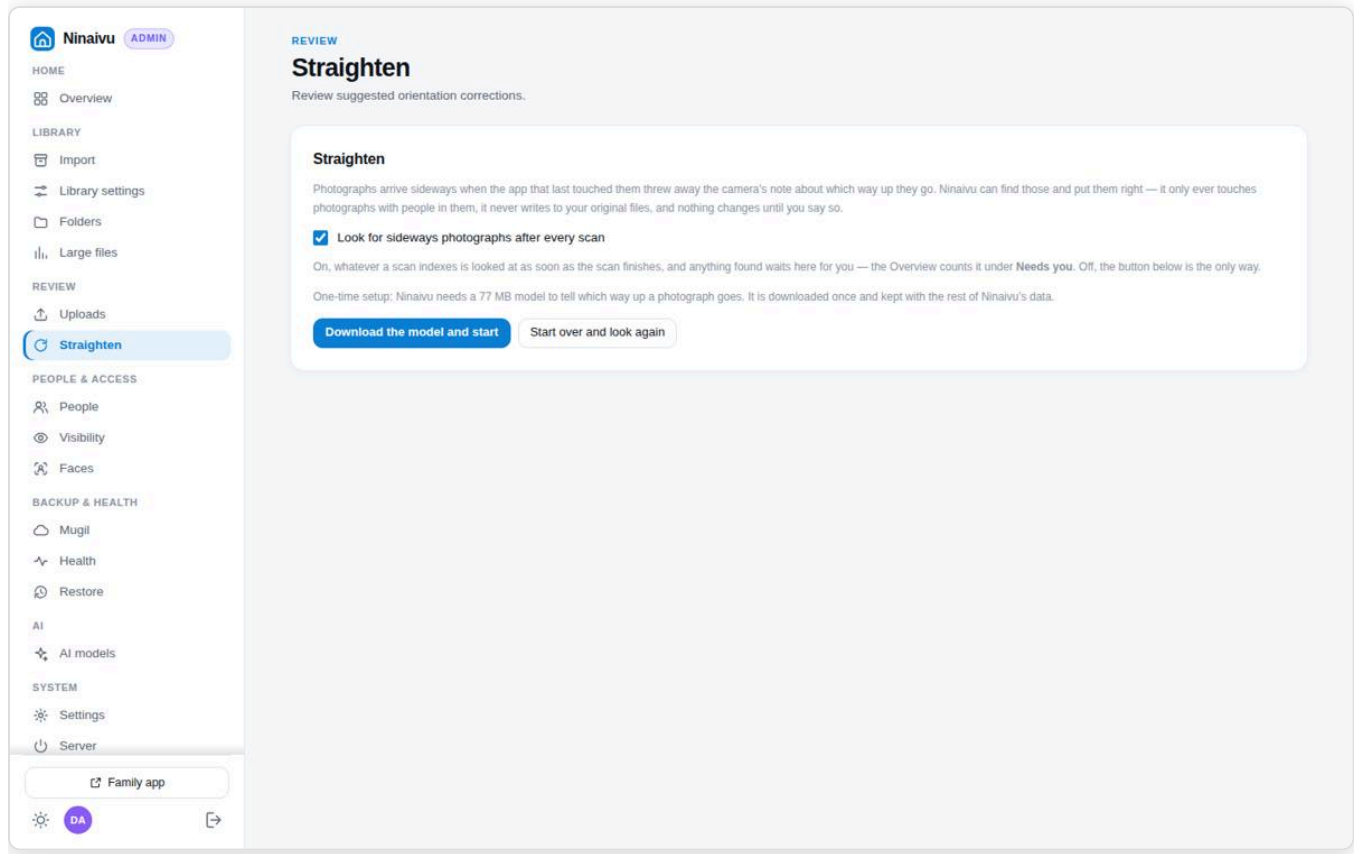
Review

Uploads



Photographs the family sent from their phones, and edits saved in Sudar, waiting to be filed or refused; and the phones that have sent backups. Nothing a family member sends joins the library without passing here, unless you tick **Trust phone backups**; an administrator's own phone skips the review.

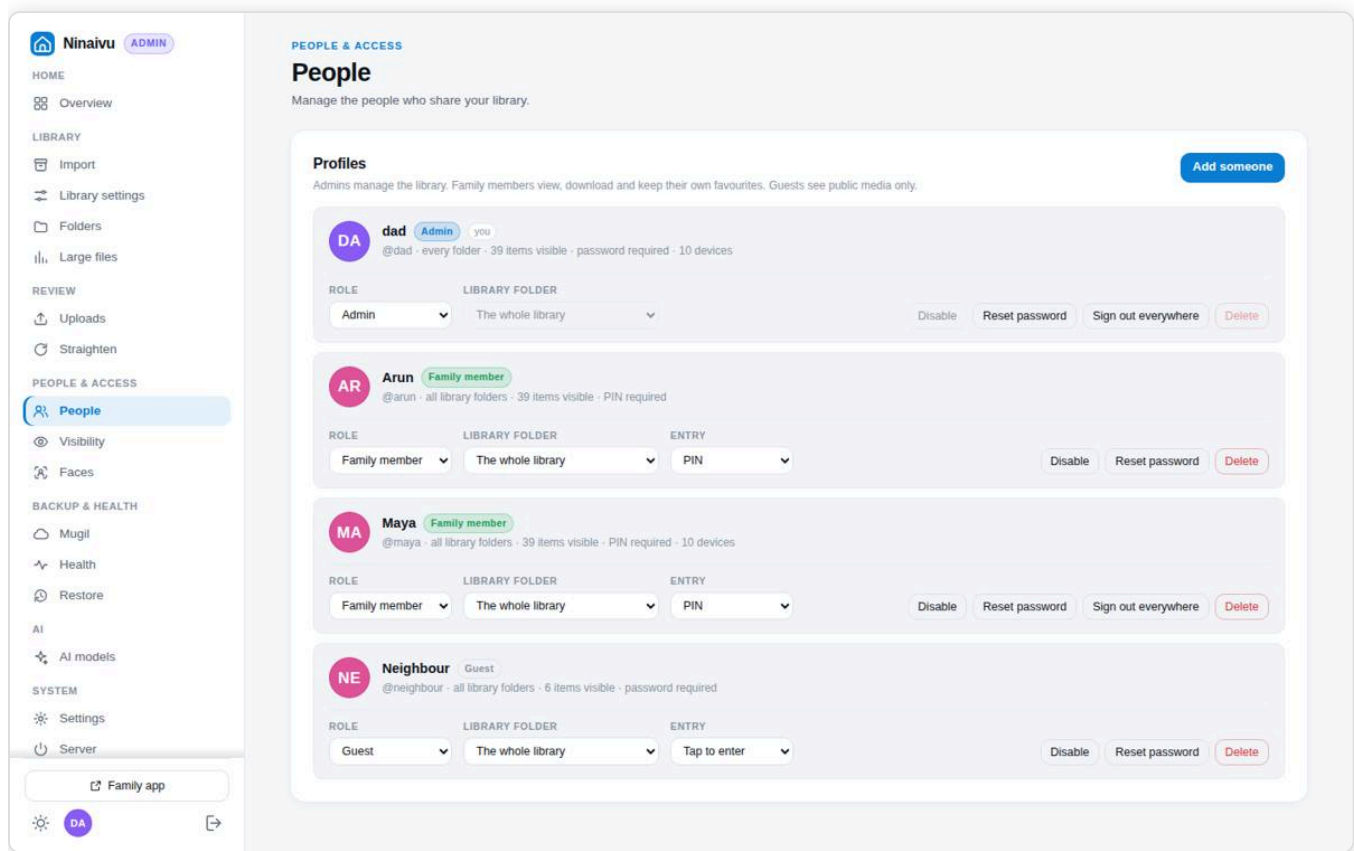
Straighten



Photographs the model thinks are on their side, strongest guess first, each shown the way it would be left. Click one to skip it; **Straighten them** turns the rest in one batch. It only ever suggests a quarter turn, and by default only for photographs with a person in them — a found face is a second witness that this way up is the way a person stands. By default the looking happens by itself after every scan, over what the scan indexed, and what it finds is turned straight away, as one batch. **Undo** puts back the last batch exactly, thumbnails included. Two switches at the top change this: one turns the automatic looking off, leaving the button; the other keeps what it finds here for approval instead of turning it. A search started with the button always waits for you.

People & access

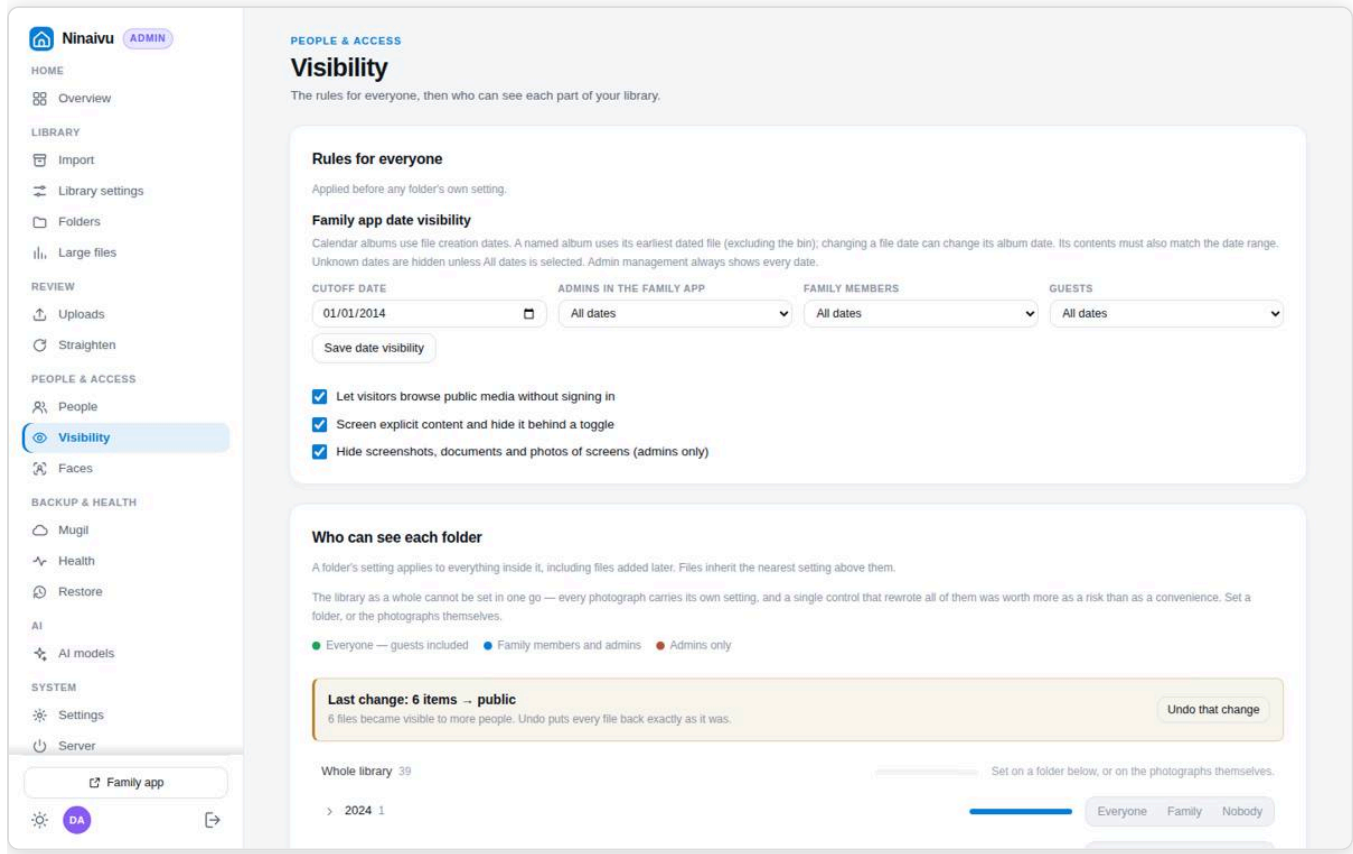
People



The household's profiles: one administrator, family members, guests. Each has a role, a way in (a PIN, a password, or a tap), and optionally a folder of their own — any library folder, or any folder inside one. Assignment and visibility stack, and the narrower always wins: a guest given a folder still sees only what is public in it.

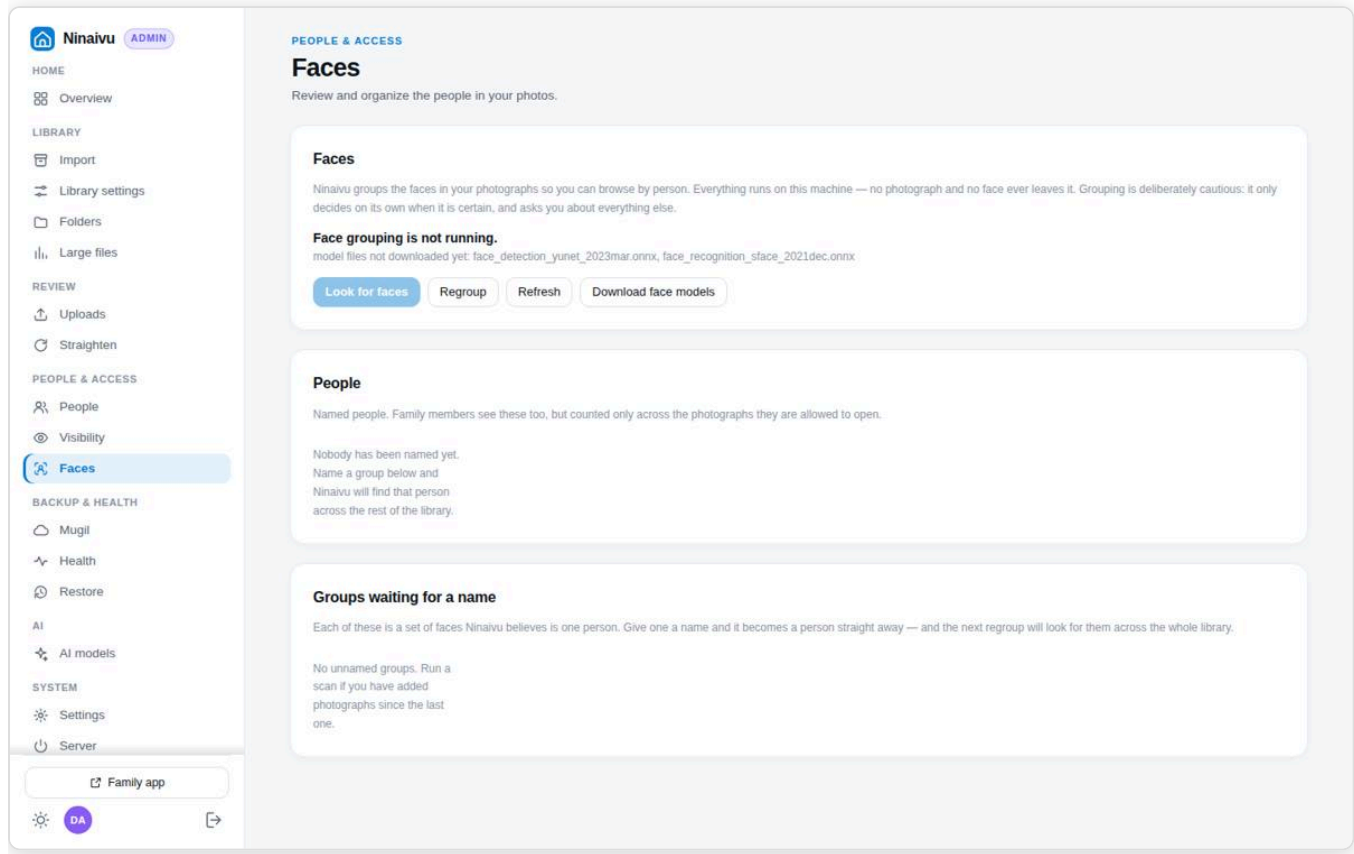
Disable is reversible and keeps the person's favourites; **delete** is final and removes the profile, its favourites and every open session. Neither touches a photograph. You cannot delete yourself or the last administrator. Disabling a profile or resetting its password ends its sessions everywhere at once.

Visibility



Who sees what. The rules for everyone come first — whether visitors may browse without signing in, whether explicit content is screened, how far back in time each role may look — and then each folder's own setting: everyone, family, or admins only. New media is *family* until somebody says otherwise, so nothing reaches a guest by accident. A folder's setting is remembered as a rule, so files scanned into it later inherit it. Setting a folder rewrites everything beneath it, so the previous state is written down first and every change can be undone in one step.

Faces



The people the face matcher found, grouped. Name one face and the rest of its group follow; groups waiting for a name are listed for a quick pass. Faces smaller than 50 pixels are skipped on purpose, which is the trade that keeps the ones it shows reliable. Faces are found and matched on this computer; the names travel only inside Mugil's encrypted copy of the index.

Backup & health

Mugil

The screenshot shows the 'Mugil' configuration page under the 'BACKUP & HEALTH' section. The left sidebar contains navigation links: HOME, Overview, LIBRARY (Import, Library settings, Folders, Large files), REVIEW (Uploads, Straighten), PEOPLE & ACCESS (People, Visibility, Faces), BACKUP & HEALTH (Mugil, Health, Restore), AI (AI models), and SYSTEM (Settings, Server). The main content area is titled 'Mugil' and describes it as 'the encrypted copy of the library in the cloud, and how it is going.' It features a 'Mugil — cloud backup' section with a 'one way - Google Drive' tag, explaining that files are uploaded once and are never cleared by Google. Below this is a 'Google account' section with a 'Needs setup' tag, detailing the one-time setup in Google Cloud. It includes a list of five steps: opening console.cloud.google.com, enabling the Google Drive API, creating an OAuth client, adding the authorized redirect URI (http://127.0.0.1:5602/api/cloud/callback), and pasting the client ID and secret. There are input fields for 'Client ID' and 'Client secret', and a 'Save' button. A 'Connect Google account' button is also present. A note at the bottom states: 'Ninaivu asks for the narrowest permission Google offers — access to the files it creates itself, and nothing else in your Drive.'

Mugil

Mugil: the encrypted copy of the library in the cloud, and how it is going.

Mugil — cloud backup one way - Google Drive

A second copy of the library, kept in one folder in a Google Drive account you choose. It only ever goes out.

What that means, exactly

- Every file is uploaded **once**. The record of what has gone is kept here and is never cleared by anything Google does.
- Deleting a photograph in Drive does **not** delete it here, and does not cause it to be uploaded again.
- Uploading never changes or removes anything on this machine. Getting files back is a separate step, *Restore from Google Drive* below, and it never overwrites a file.
- Ninaivu works inside **one folder** it makes itself. The rest of the Drive account is not read and not touched.
- Hidden items stay here unless you say otherwise.

Google account Needs setup

One-time setup in Google Cloud

Ninaivu cannot ship a Google client secret — a secret inside an app anybody can unzip is not a secret, and a shared one would put every household behind one quota. It is about five minutes:

1. Open console.cloud.google.com and make a project.
2. Enable the [Google Drive API](#) for it.
3. Under *Credentials*, create an OAuth client of type **Web application**.
4. Add this exact authorised redirect URI: <http://127.0.0.1:5602/api/cloud/callback> Copy
5. Paste the client ID and secret below.

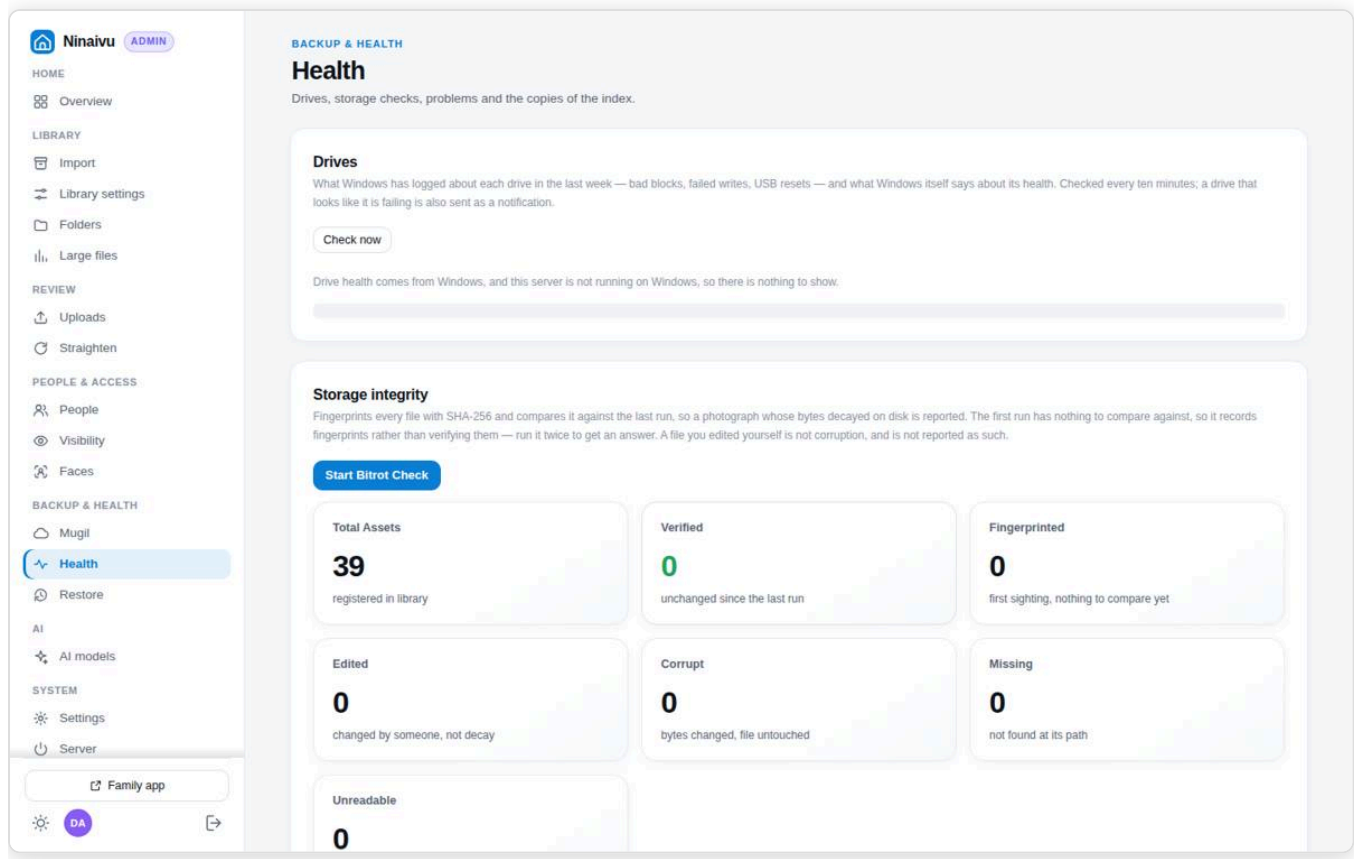
Client ID (apps.googleusercontent.com) Client secret Save

Connect Google account

Ninaivu asks for the narrowest permission Google offers — access to the files it creates itself, and nothing else in your Drive.

Mugil (முகில், *cloud*): the encrypted copy of the library on Google Drive — the account, the encryption key and its recovery file (nothing is uploaded until the key is made), what is sent and what is kept back, speed and hours, and **Test restores**: the weekly test that the copy restores, with its history. More on backup.

Health



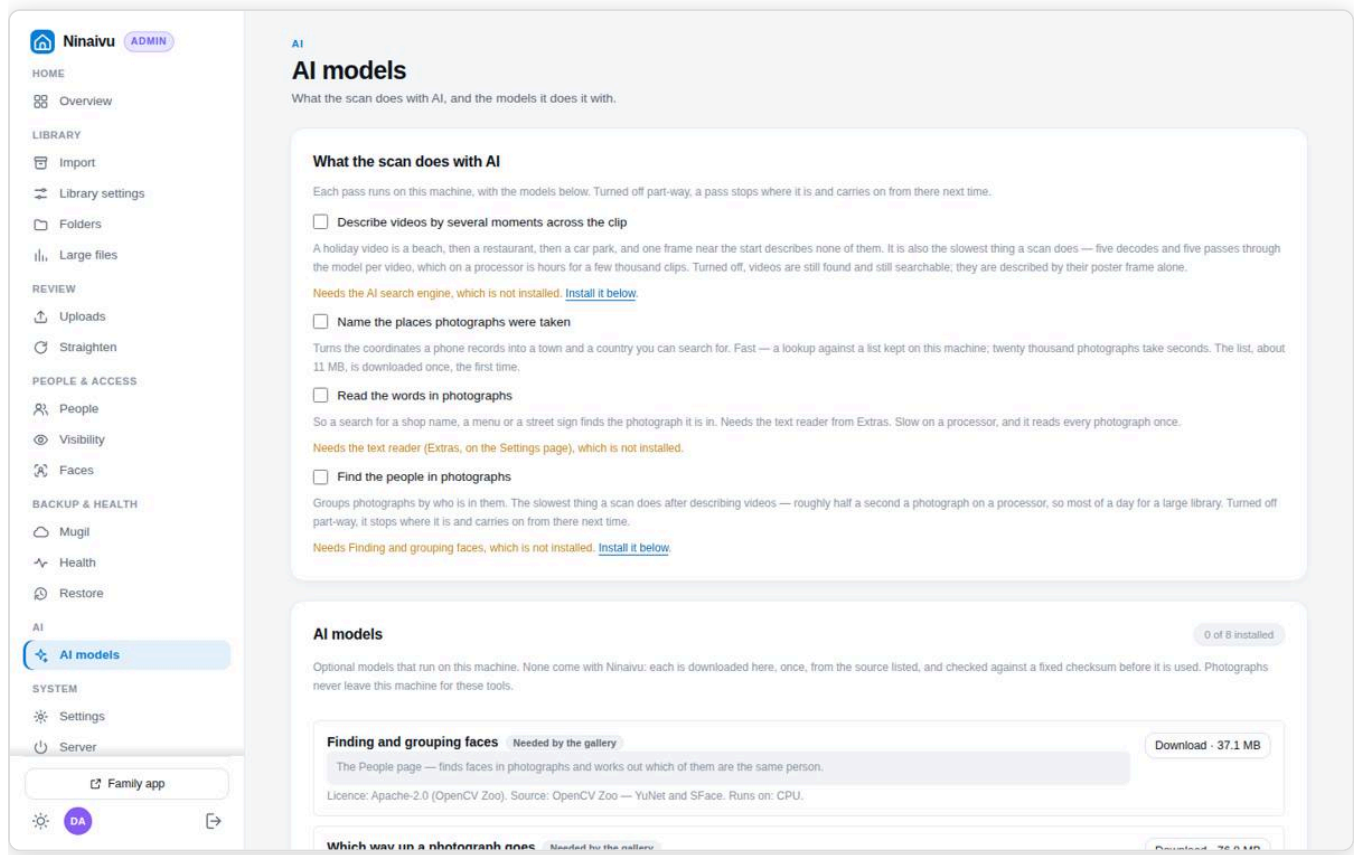
The drives and their SMART state, the storage check that reads every file back against its fingerprint, anything that went wrong since the last start, the copies of the index kept on this computer, and how the household is told when something needs them — by email or a webhook.

A photograph, once a week: on the morning you choose, Ninaivu picks the best photograph from that day in a past year and emails it to the family; a day with nothing worth sending stays quiet.

Restore

Bringing photographs back from the cloud copy, in three steps: what to bring back, where to put it, and the encryption key. Nothing here ever overwrites a file that is already there.

AI models



What the scan does with AI — naming places, reading text, finding faces, describing videos by several moments — each switch saying which model it needs and whether that model is here, then the models themselves with their downloads. Everything on this page runs on this machine. More on AI.

AI server

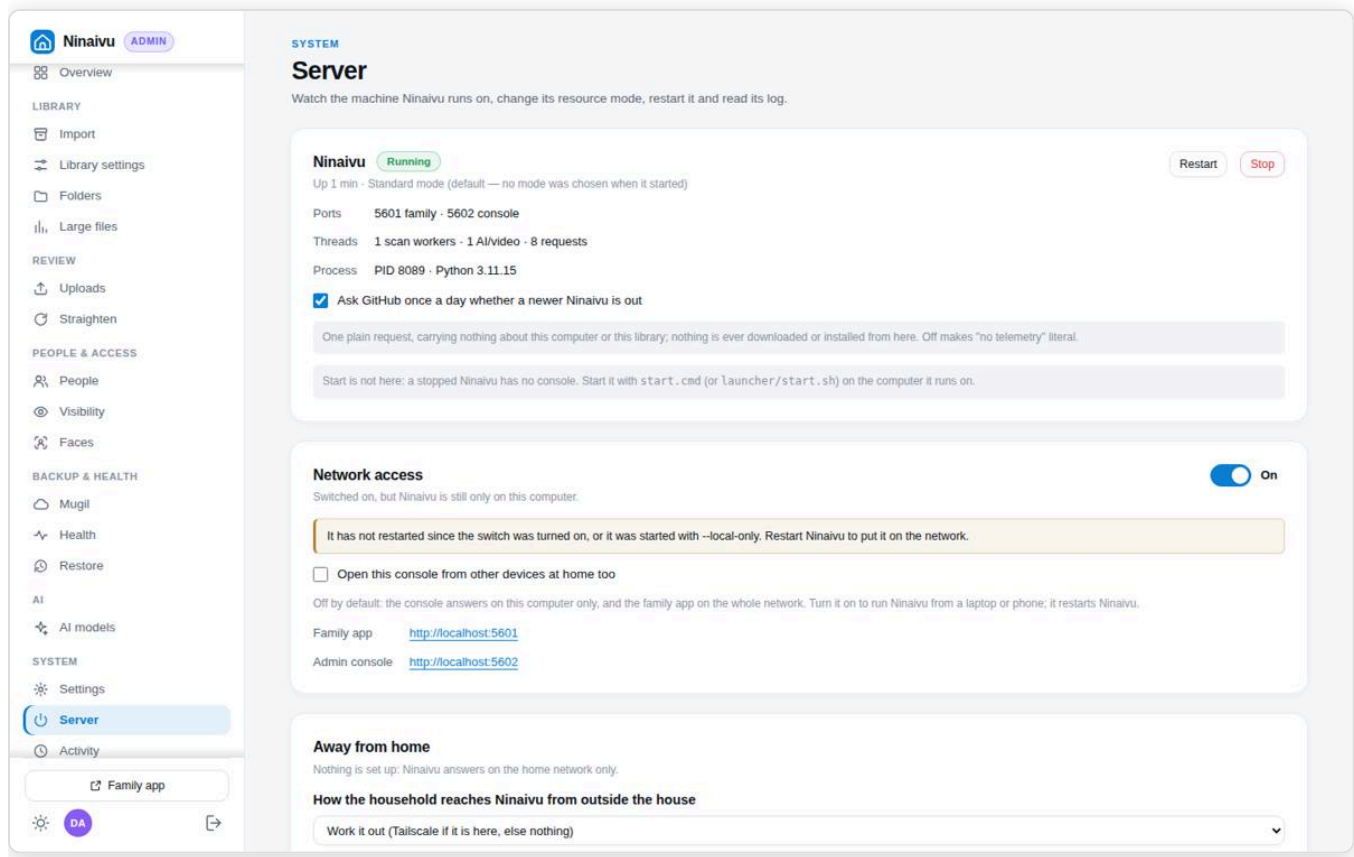
For a household with a second, stronger computer running ComfyUI: hand the heavy jobs to it. This page belongs to the Creative Studio extension and appears only while it is on.

System

Settings

This home's name and the version of Ninaivu; the **extensions** installed, each saying on its switch what it does when it is on, and whether family members may send photographs to one that goes outside the house; and **Extras**, the optional packages — search by description among them.

Server



Whether Ninaivu is running and where it answers, whether a newer version is out, the addresses to give the household, **Away from home** — how the household reaches Ninaivu from outside (remote access) — **Network access** and whether the console may be opened from other devices at home too, the resource mode (standard, performance, power-saving), a restart, the log, and the HTTPS certificate. The ports are shown here; they are set when Ninaivu starts (`--port` , `--admin-port`).

Activity

What is running now and who is using the computer, so the backup and the scans can make way for the household. **Balanced** gives way to video playback; **quiet** waits whenever anyone is using the app; **overnight** keeps the heavy work for the night hours and runs it flat out then.

Performance

The screenshot displays the 'Performance' section of the Ninaivu application interface. On the left is a sidebar with navigation options: Import, Library settings, Folders, Large files, REVIEW (Uploads, Straighten), PEOPLE & ACCESS (People, Visibility, Faces), BACKUP & HEALTH (Mugil, Health, Restore), AI (AI models), and SYSTEM (Settings, Server, Activity, Performance). The 'Performance' option is currently selected. Below the sidebar, there are buttons for 'Family app' and a user profile icon labeled 'DA'.

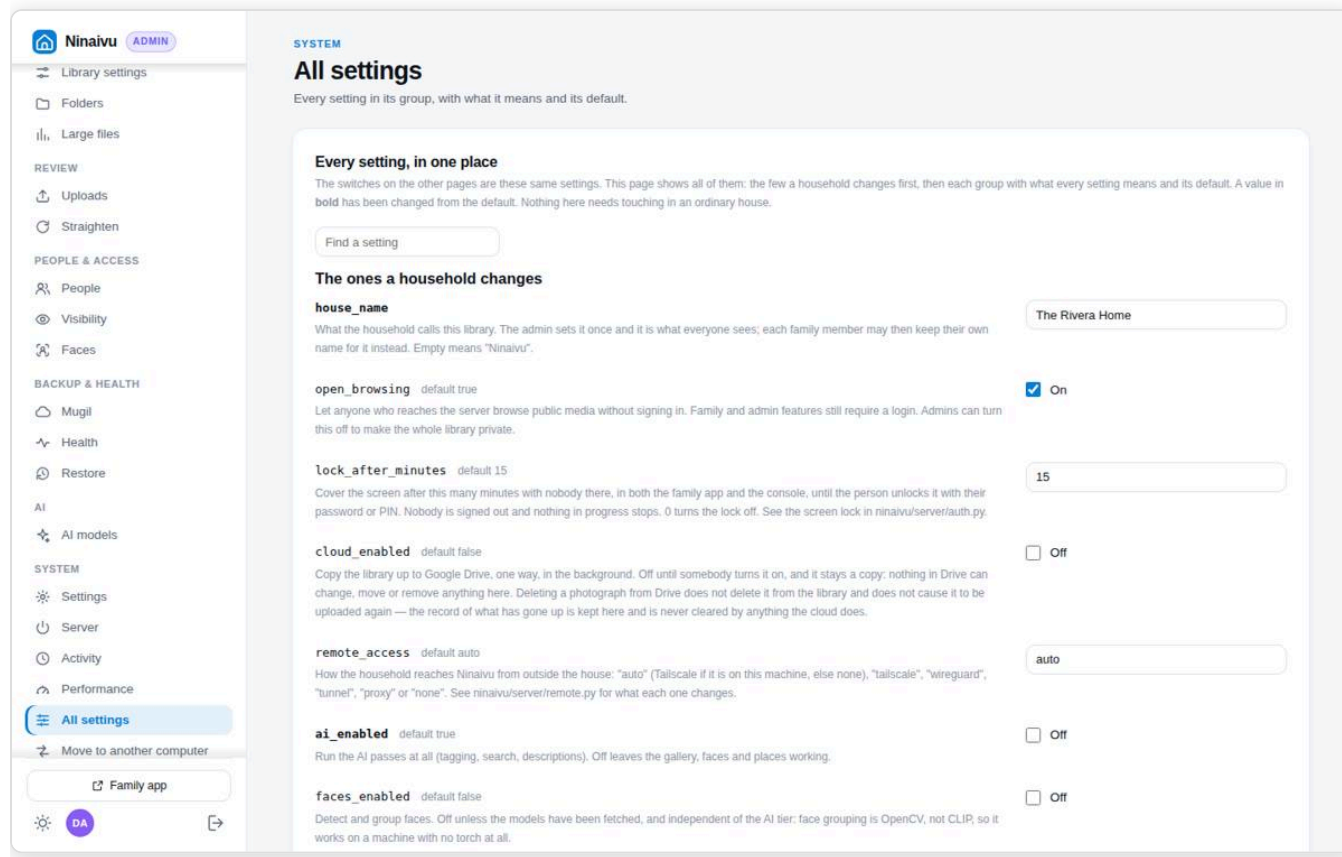
The main content area is titled 'SYSTEM Performance' with the subtitle 'What this computer can do for Ninaivu, and what would help it do more.' It features several informational cards:

- What would help:** A card indicating the system configuration: 'Intel(R) Xeon(R) Processor @ 2.10GHz - 2 cores - 7.8 GB - Linux 6.18.44-1c-v49 — Nothing to change.' It includes a 'Measured 11:21:00 AM' timestamp and a 'Check again' button. Below this, a green bar states 'FINE Enough memory' with the detail '7.0 GB available; Ninaivu uses 0.1 GB.'
- This computer:** A section with five sub-cards:
 - Kind of computer:** 'Basic' with a note: 'no graphics processor, and the image model is not installed (Settings → Extras) — the image model, descriptions and text reading are not attempted here'.
 - Processor:** 'Intel(R) Xeon(R) Processor @ 2.10GHz' with '2 cores'.
 - Memory:** '7.8 GB' with '7.0 GB free now - Ninaivu uses 0.1 GB'.
 - Graphics:** 'None' with the note 'None Ninaivu can use'.
 - Busy now:** '9%' with 'processor in use'.
- How Ninaivu uses it:** A table showing current settings:

Resource mode	standard — 2 workers, 1 AI threads, 8 web threads
Background work	Balanced — alongside the household
Image model	Off
Waiting for analysis	0 photographs and videos

What this computer can do for Ninaivu — which kind of computer it is (Basic or Full), the processor, memory, graphics, drives, how fast the last scans went, what is waiting — and what would help it do more.

All settings



Every setting Ninaivu has, in six groups, with what each means and its default. The switches on the other pages are these same settings. Nothing here needs touching in an ordinary house.

Move to another computer

Moving Ninaivu, or telling it the library has moved, without re-indexing everything. The details.

Running it from the command line

The tray and the installers start Ninaivu for you. From a checkout, `python -m ninaivu <folder>` does, and these are the flags worth knowing:

Flag	
<code>--admin USER:PASSWORD</code>	make the administrator without a browser
<code>--port</code> , <code>--admin-port</code>	the two ports (80 — 443 with HTTPS — and 3000 by default)
<code>--local-only</code>	this computer only; phones cannot reach it
<code>--https</code> , <code>--cert</code> , <code>--key</code>	HTTPS with Ninaivu's own certificate, or yours
<code>--ai auto clip light off</code>	the image model; <i>auto</i> follows the hardware tier
<code>--faces</code> , <code>--ocr</code> , <code>--places</code>	turn the passes on from the start
<code>--no-watch</code>	do not watch the folders for changes
<code>--open-browsing</code> / <code>--private</code>	whether visitors may look without signing in
<code>--lock-roots</code>	confine the folder picker to the library folders
<code>--admin-host 0.0.0.0</code>	open the console to the network from the start
<code>--strict-port</code>	refuse to start rather than move to another port
<code>--name</code> , <code>--no-mdns</code>	the name on the home network (<code>ninaivu.local</code>), or none
<code>--map-tiles</code>	OpenStreetMap tiles on the map (they show roughly where photographs were taken)

And the maintenance commands, which work in an installed copy too:

Command	
<code>ninaivu backup --out <folder></code>	a copy of the index, settings and certificate
<code>ninaivu list-backups <folder></code>	the copies in a folder
<code>ninaivu restore <file></code>	put one back (Ninaivu stopped)
<code>ninaivu reroot --from <old> --to <new></code>	tell the index the library moved

Some settings also read an environment variable (`NINAIVU_STATE_DIR` , `NINAIVU_AI_ENGINE` , `NINAIVU_HARDWARE_TIER` , `NINAIVU_FACES` , ...; the full list is in `server/config.py`). Folders and ports from the environment always apply; anything about behaviour only fills in what nobody chose in the console. The state directory (`~/.ninaivu` , or `$XDG_DATA_HOME/ninaivu` , or `NINAIVU_STATE_DIR`) holds the index, the thumbnails and `config.json` ; the AI models live in `.ai-models` beside the application, or `ai_models_dir` .